

Matrices of Linear Transformations

Geometric Algorithms
Lecture 8

Practice Problem

$$T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} x_2 - 3x_1 \\ 9x_1 - 3x_2 \end{bmatrix} \quad \mathbf{b} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

Is $\mathbf{b}$ in the range of T ? If so, determine $\mathbf{v}$ such that $T(\mathbf{v}) = \mathbf{b}$

$$T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} x_2 - 3x_1 \\ 9x_1 - 3x_2 \end{bmatrix}$$


$$\begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} x_2 - 3x_1 \\ 9x_1 - 3x_2 \end{bmatrix}$$

$$\begin{aligned} 2 &= x_2 - 3x_1 = -3x_1 + x_2 \\ -1 &= 9x_1 - 3x_2 \end{aligned}$$

$$\begin{bmatrix} -3 & 1 & 2 \\ 9 & -3 & -1 \end{bmatrix} \sim \begin{bmatrix} -3 & 1 & 2 \\ 0 & 0 & 5 \end{bmatrix}$$

1 2

$$0 = 5$$

so
 $\vec{b} \notin \text{ran}(T)$

Keywords

matrix of a linear transformation

standard basis vectors (standard coordinate vectors)

2D linear transformations

the unit square

one-to-one

onto

Geometry of Matrix Transformations

Motivating Questions

What kind of functions can we define this way?

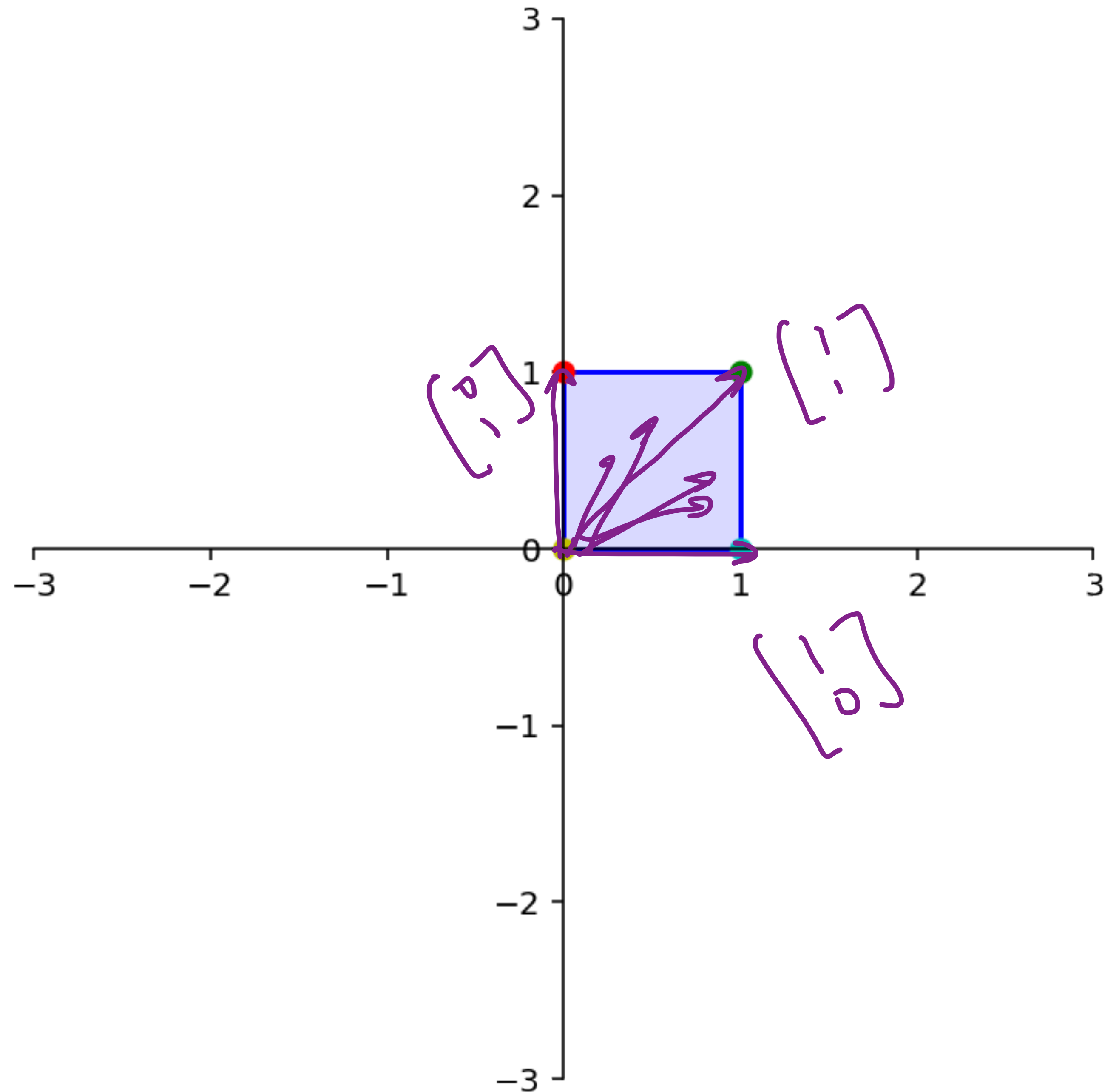
How do we interpret what the transformation does to a set of vectors?

How does this relate to matrix equations?

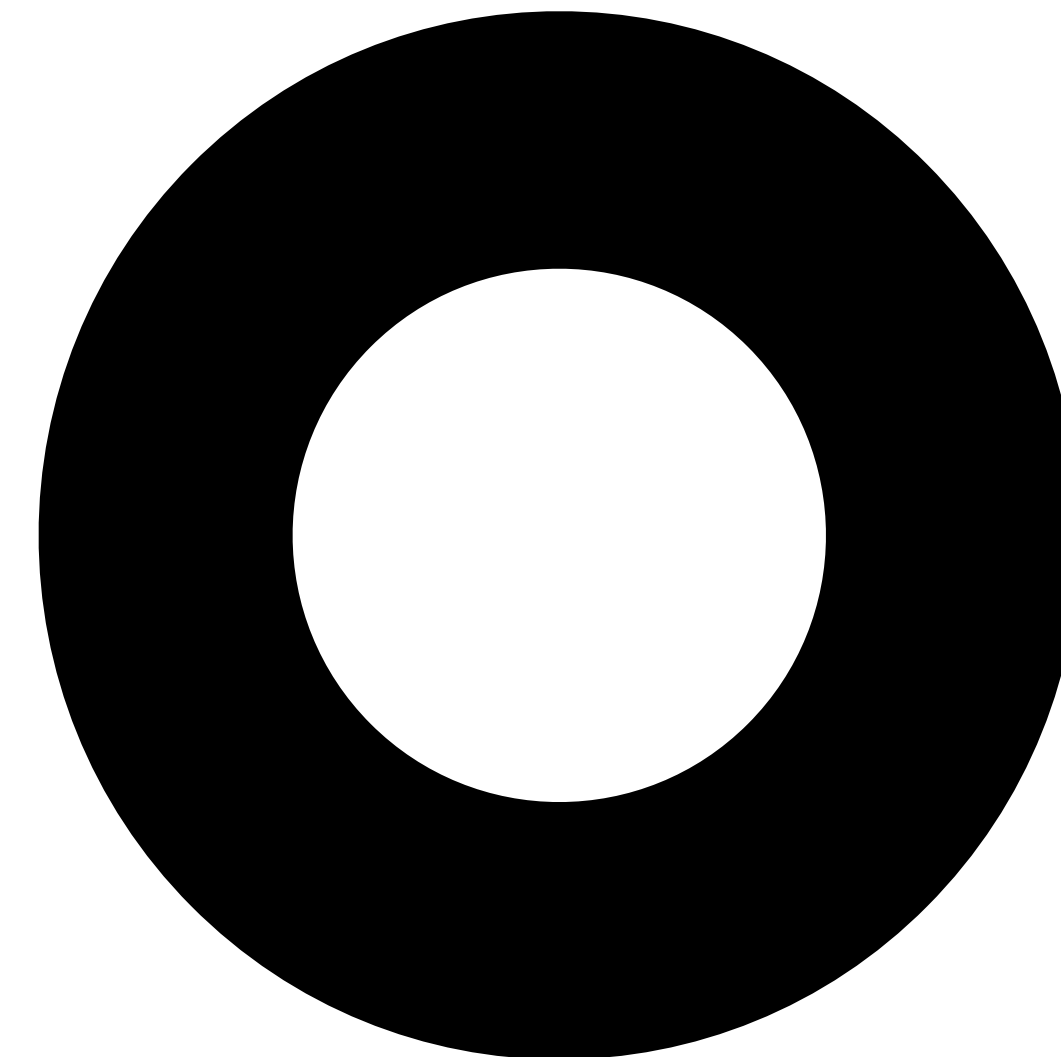
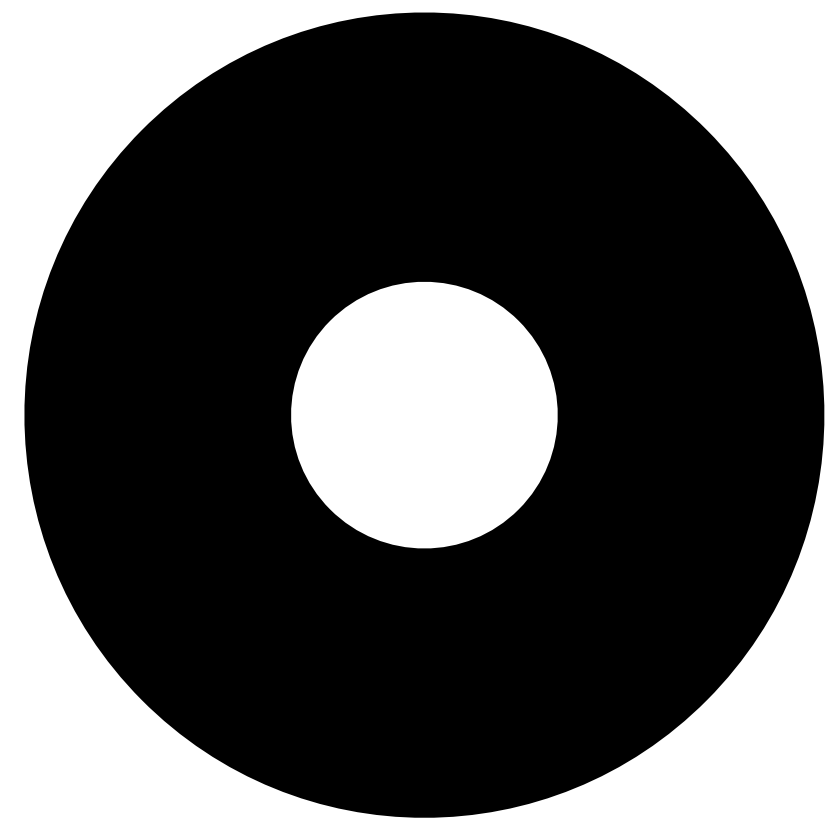
Matrix transformations change the
"shape" of a set of set of
vectors (points)

The Unit Square

The *unit square* is the set of points in $\mathbb{R}^2$ enclosed by the points $(0,0)$, $(0,1)$, $(1,0)$, $(1,1)$.

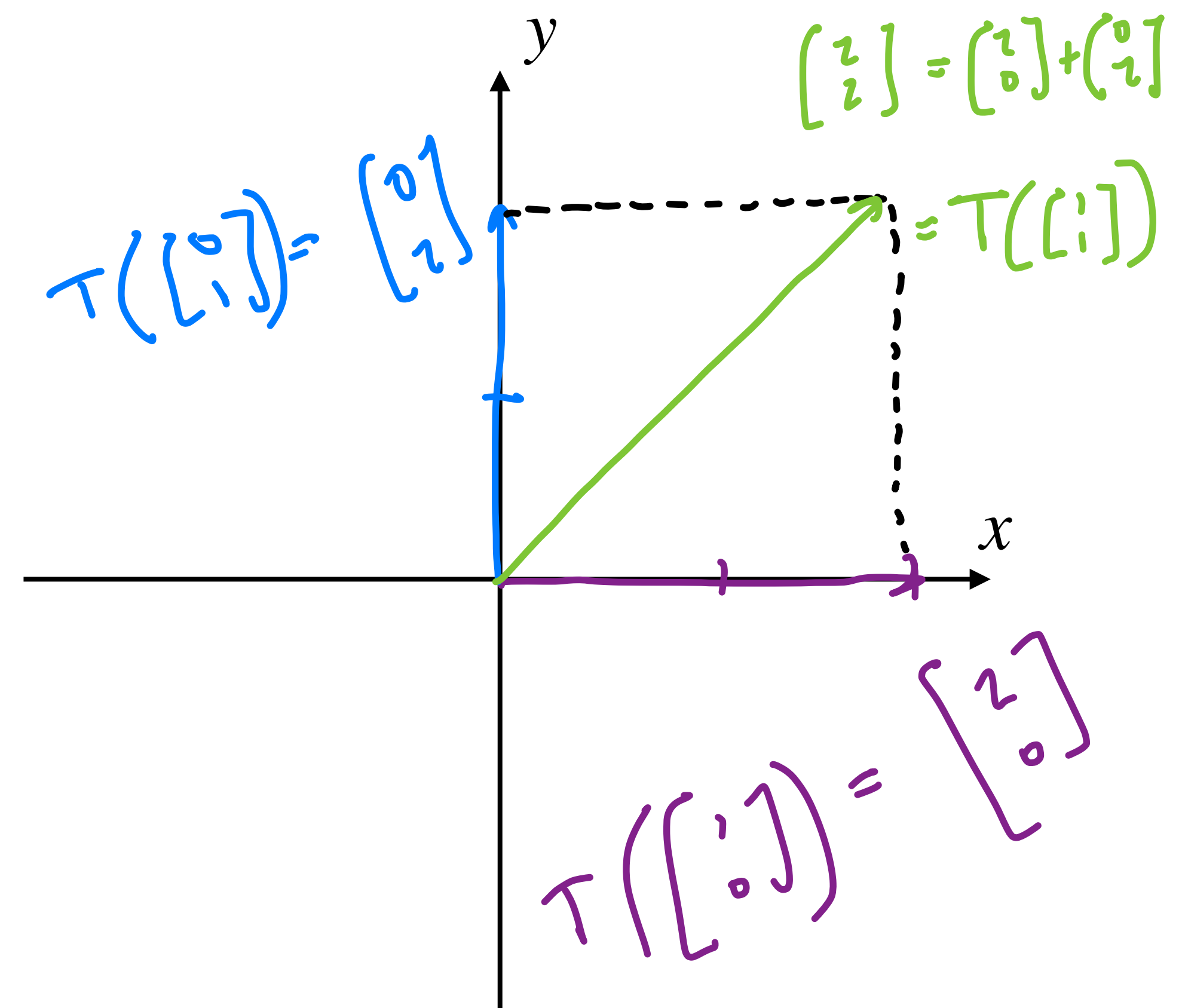
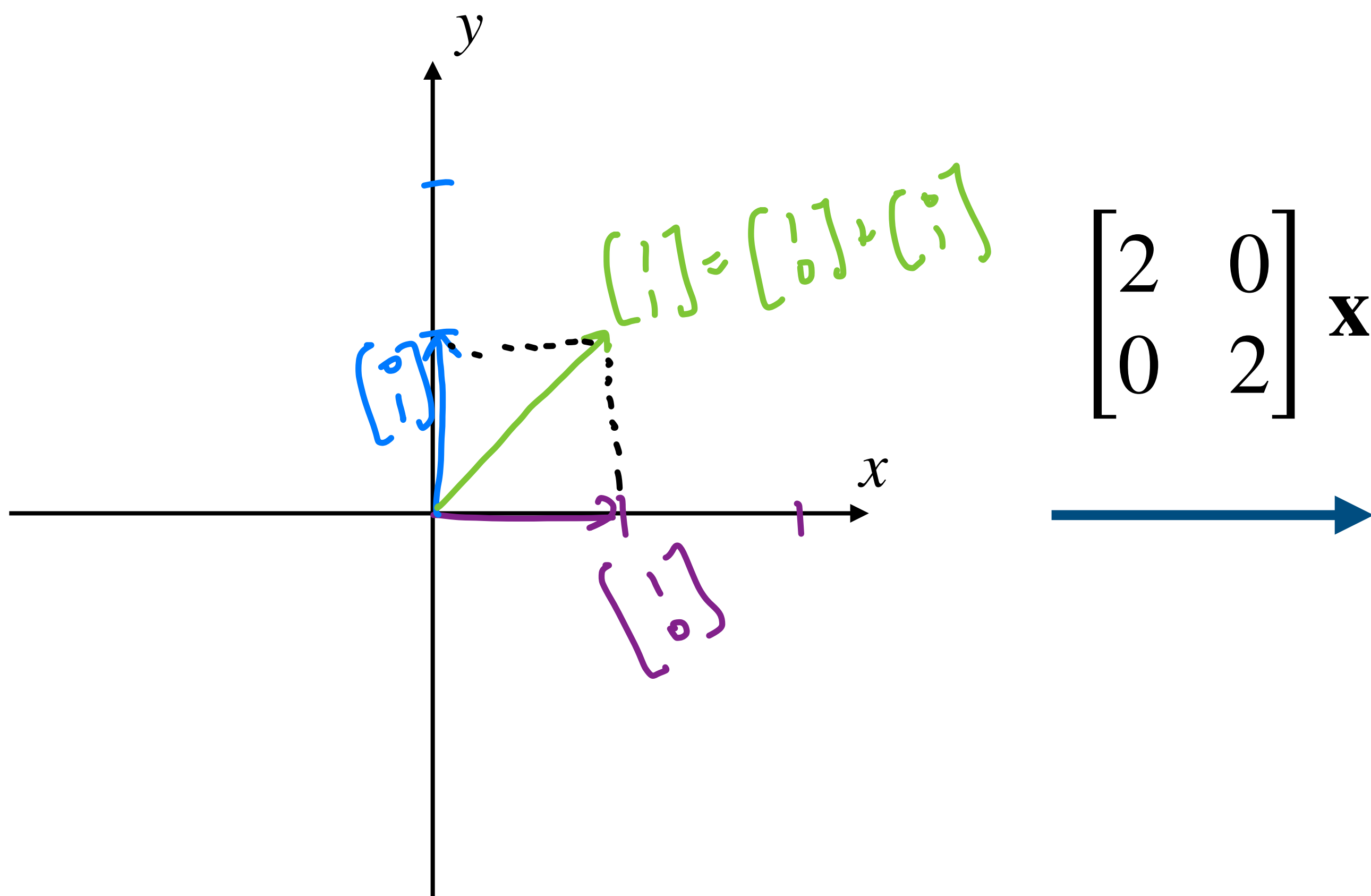


Example: Dilation



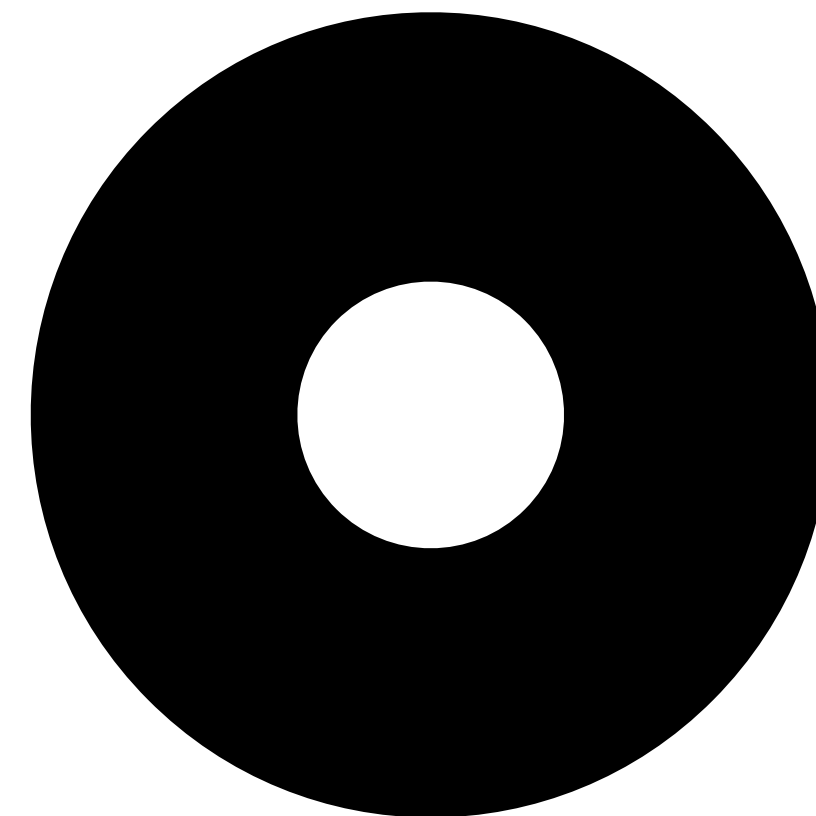
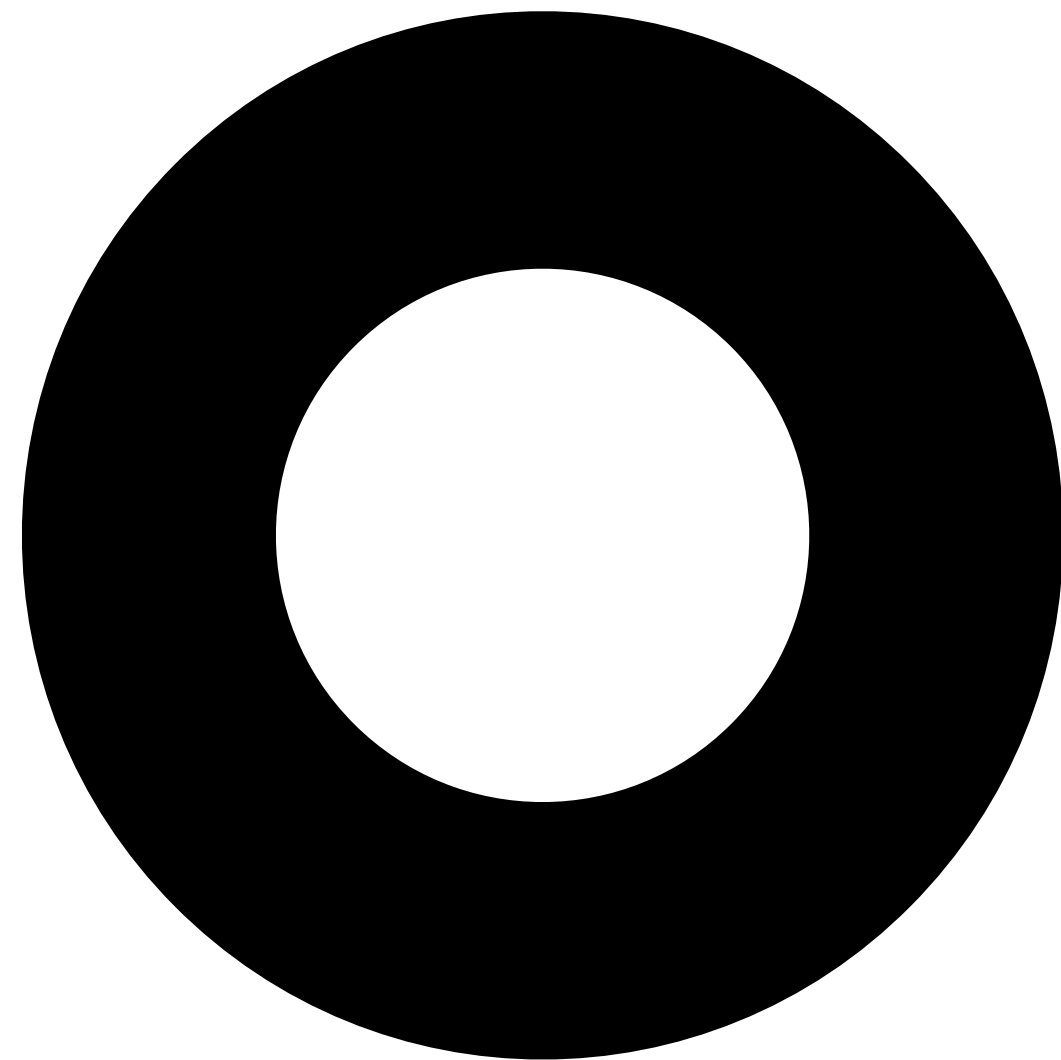
Example: Dilation

$$\begin{bmatrix} r & 0 \\ 0 & r \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} rx_1 \\ rx_2 \end{bmatrix}$$



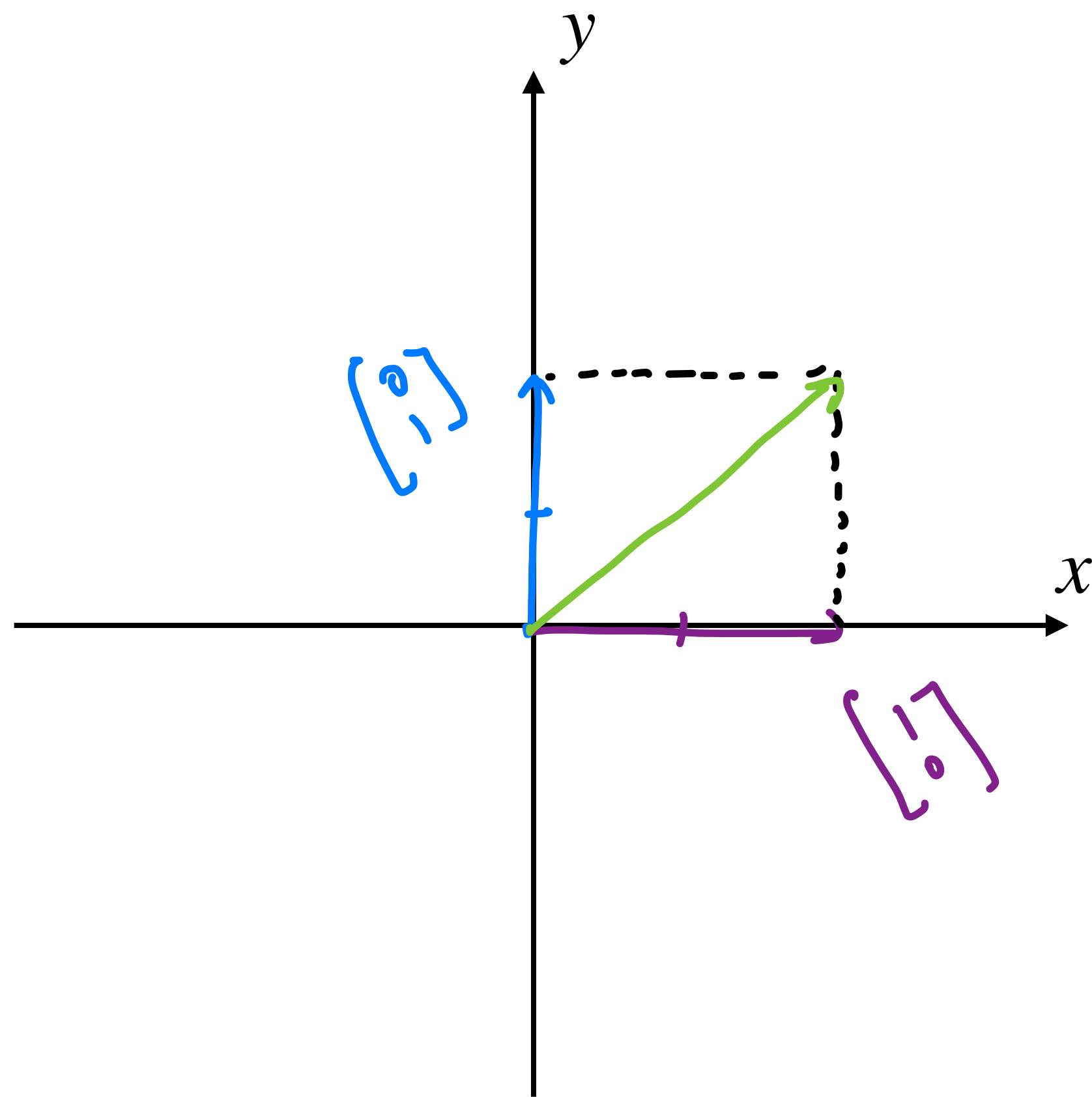
if $r > 1$, then the transformation pushes points away from the origin

Example: Contraction

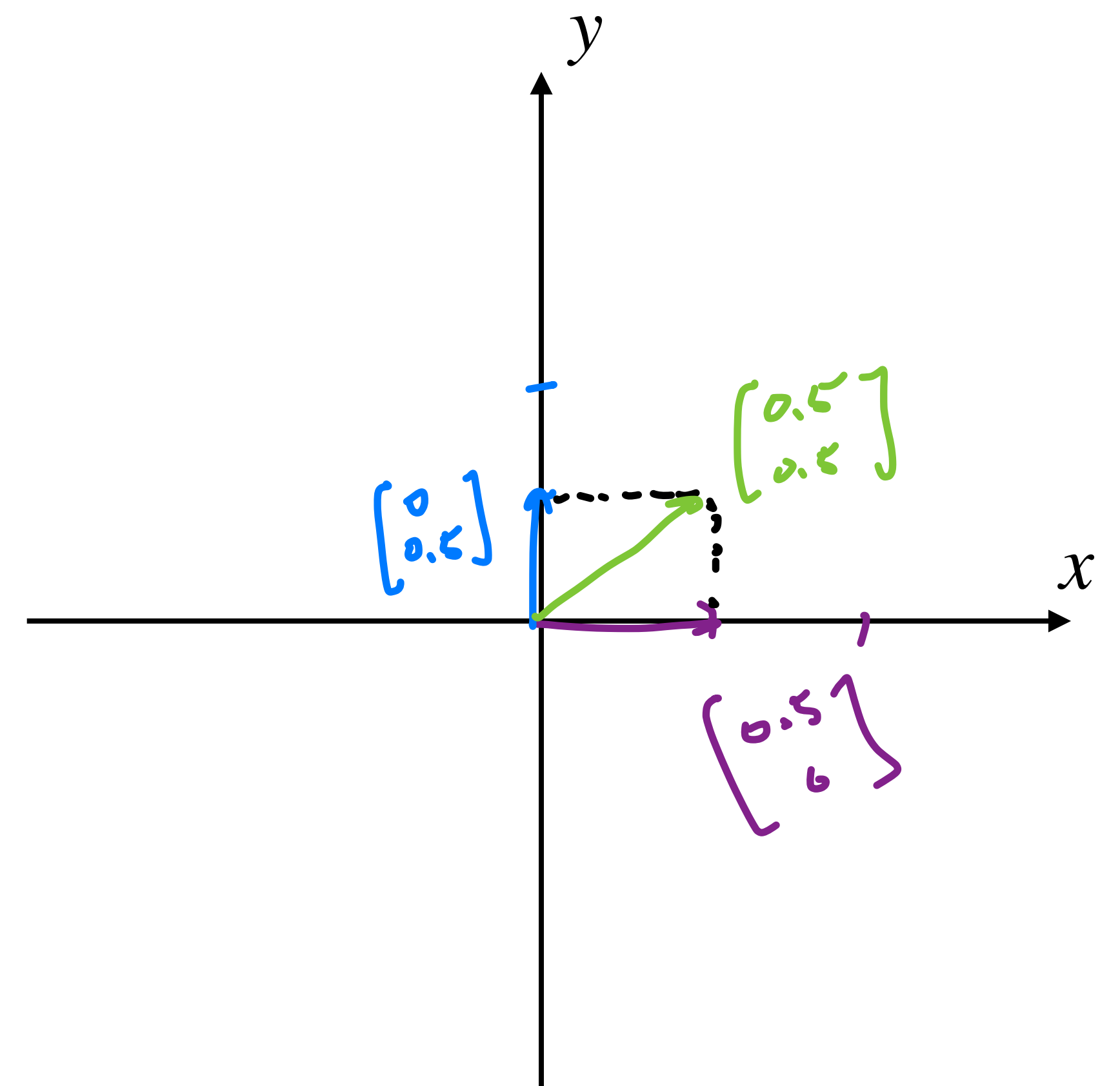


Example: Contraction

$$\begin{bmatrix} r & 0 \\ 0 & r \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} rx_1 \\ rx_2 \end{bmatrix}$$



$$\begin{bmatrix} 0.5 & 0 \\ 0 & 0.5 \end{bmatrix} \mathbf{x}$$



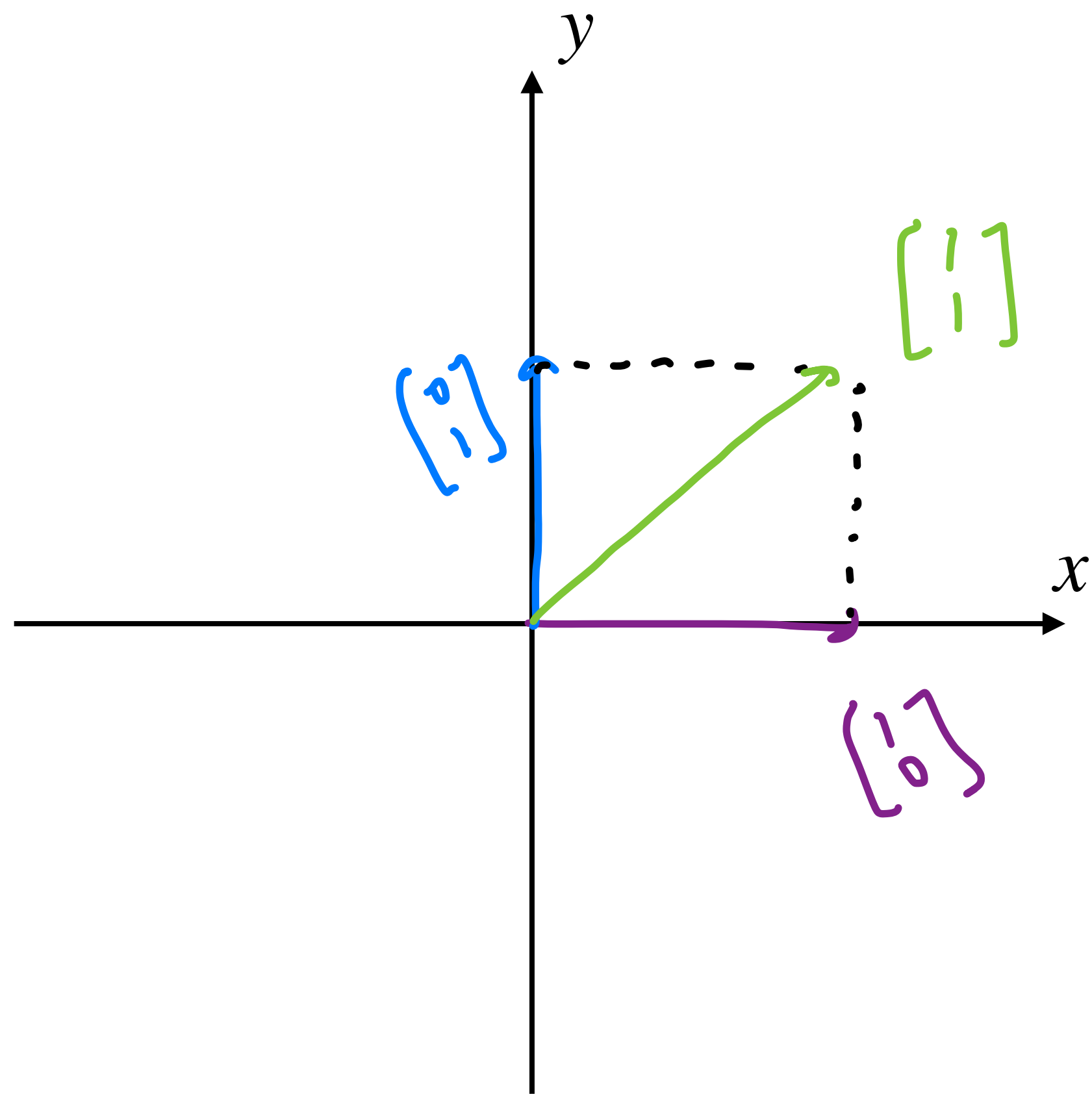
if $0 \leq r \leq 1$, then the transformation pulls points towards the origin

Example: Shearing



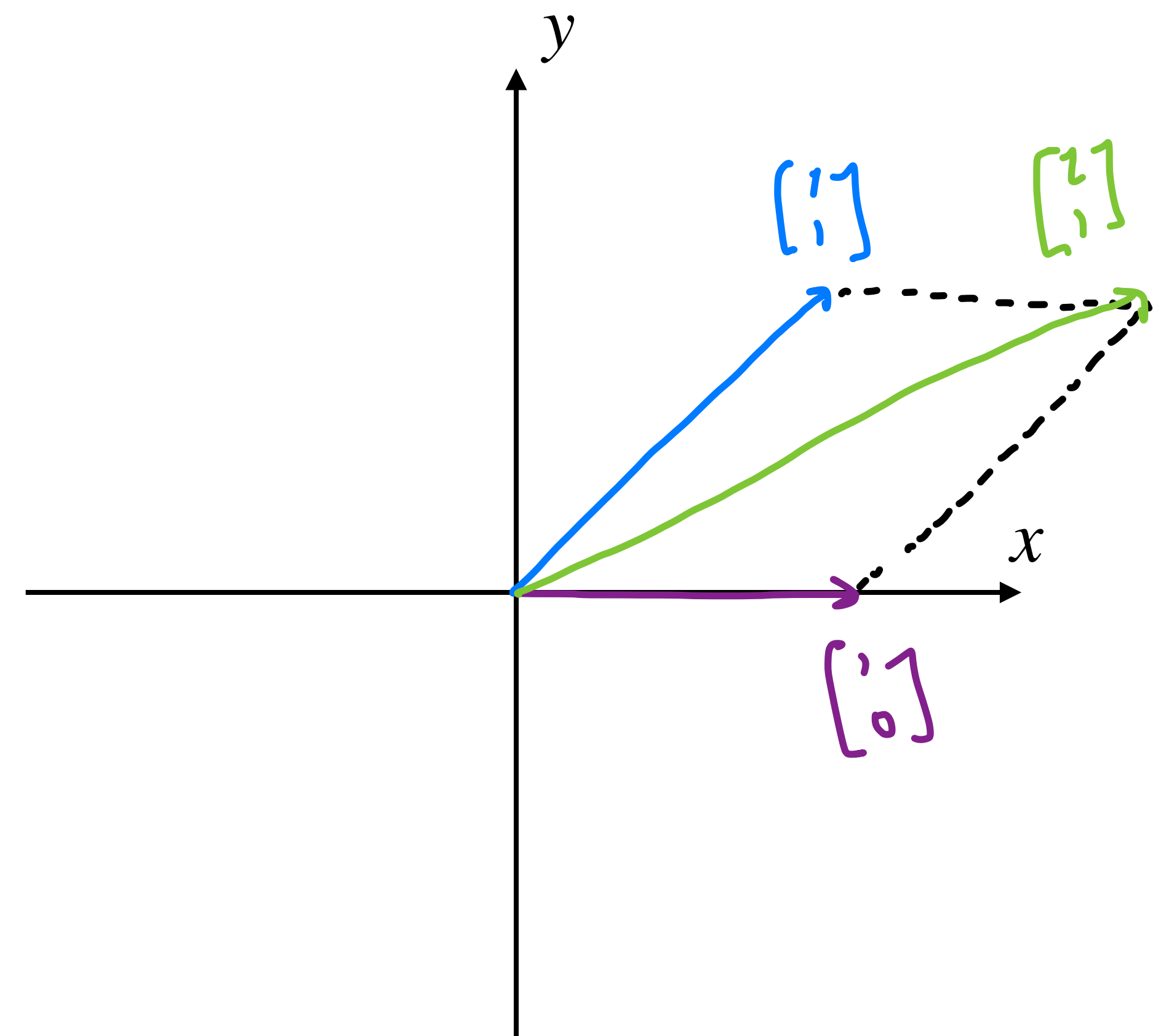
Example: Shearing

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 + x_2 \\ x_2 \end{bmatrix}$$



$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \mathbf{x}$$

A blue arrow points from the first diagram to the second, indicating the transformation.



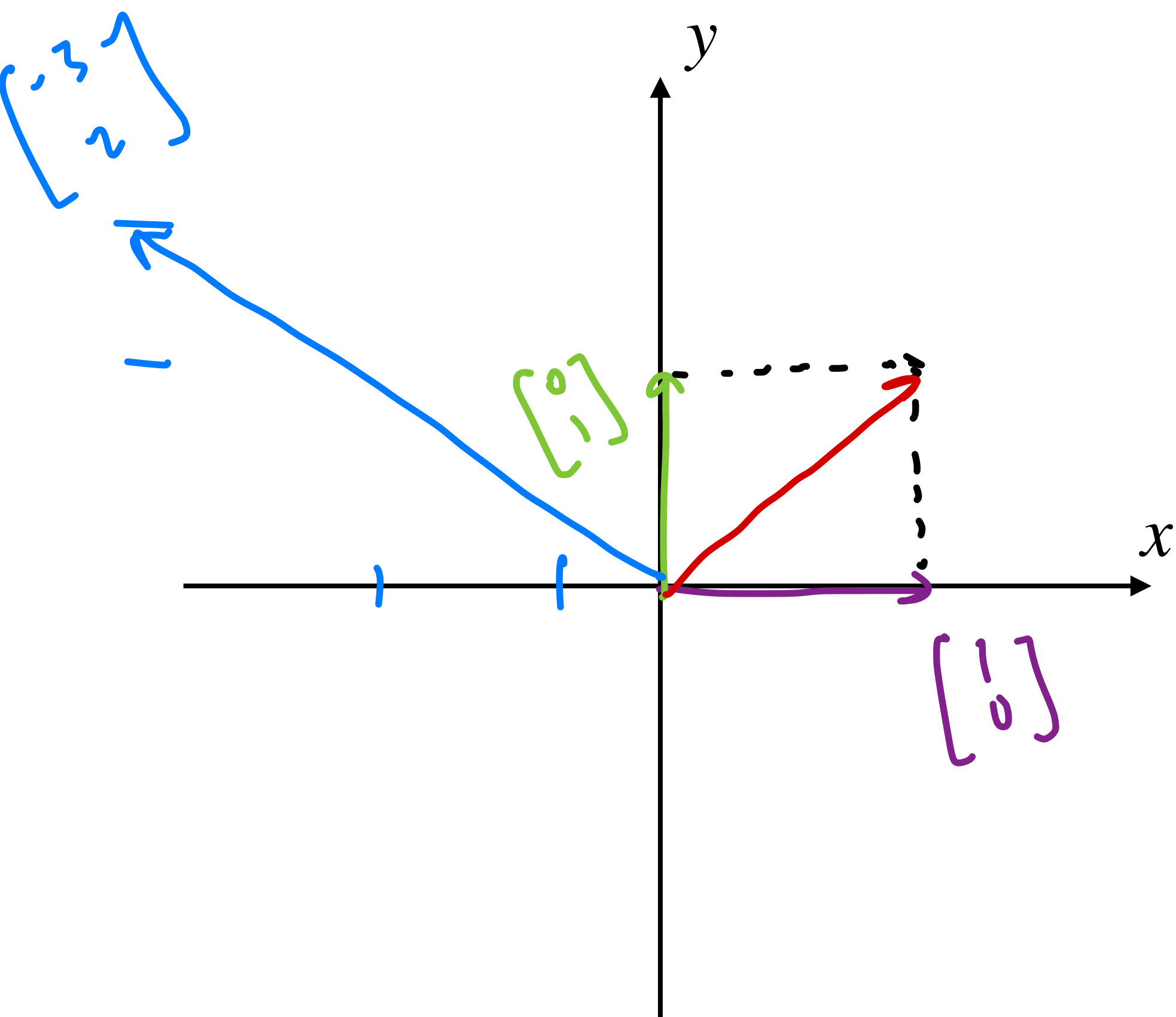
Question

$$\mathbf{x} \mapsto \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \mathbf{x}$$

Draw how this matrix transforms points. What kind of transformation does it represent?

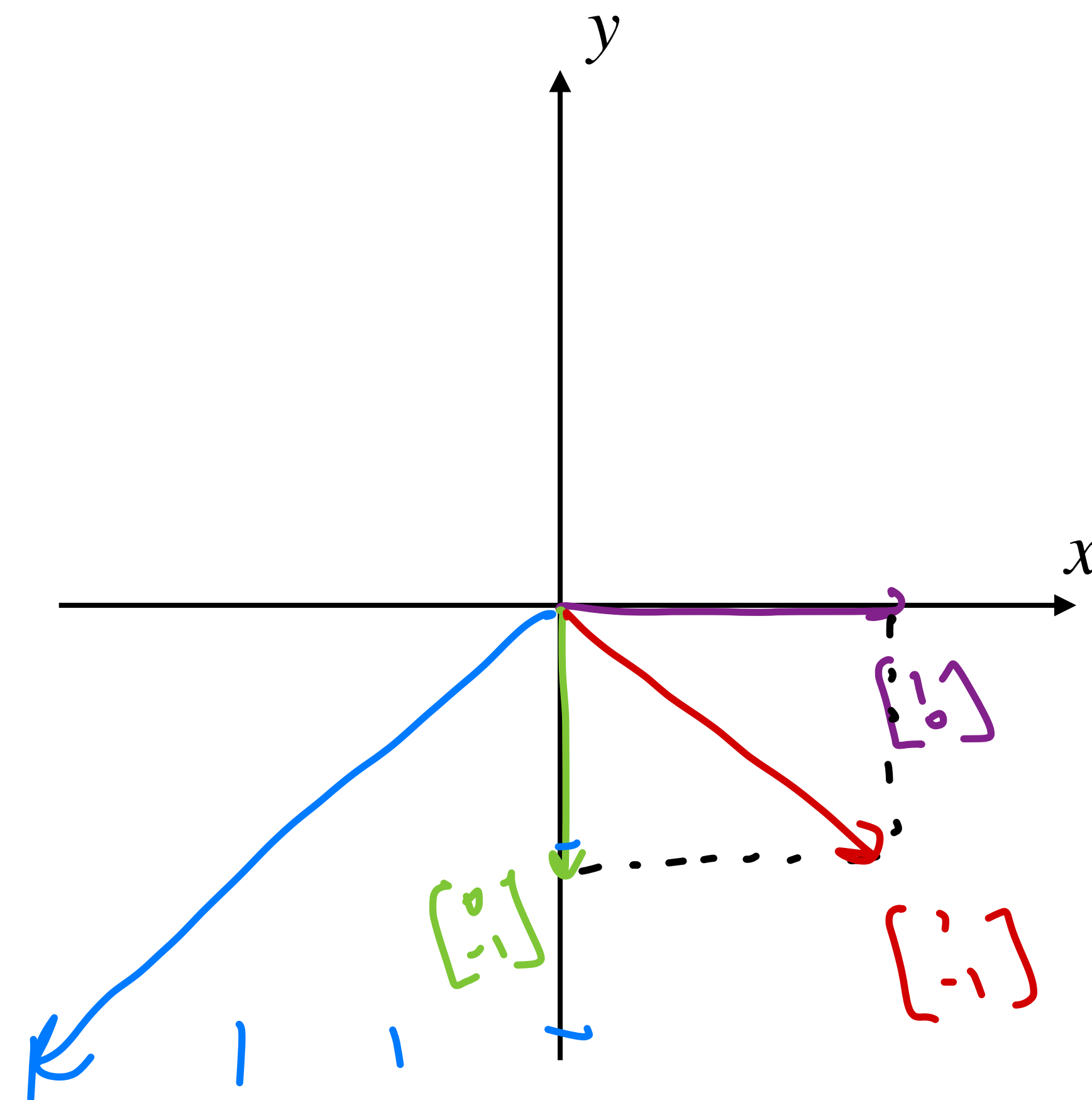
Answer: Reflection

$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 \\ -x_2 \end{bmatrix}$$



$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \mathbf{x}$$

A blue arrow points from this equation towards the right graph.



demo

Matrix of a Linear Transformations

Recall: Linear Transformations

Df. $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$

(additivity)

$$T(\vec{u} + \vec{v}) = T(\vec{u}) + T(\vec{v})$$

$$T(c\vec{v}) = cT(\vec{v})$$

(homogeneity)

$\vec{x} \mapsto A\vec{x}$ is always linear

Recall: Linear Transformations

Df. A transformation $T: \mathbb{R}^m \rightarrow \mathbb{R}^n$ is *linear* if it satisfies the following two properties

1. $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ (additivity)

2. $T(c\mathbf{v}) = cT(\mathbf{v})$ (homogeneity)

Recall: Linear Transformations

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2. $T(c\mathbf{v}) = cT(\mathbf{v})$ (homogeneity)

Matrix transformations are linear transformations

Are there any other kinds
of linear transformations?

NO

Matrix of a Linear Transformation

Thm. T is linear if and only if there is a matrix A such that $T(\mathbf{v}) = A\mathbf{v}$. whose corresponding transformation is T (which "implements" T)

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Linear transformations are **exactly**
matrix transformations

Today's Fundamental Concern

Given a linear transformation T , how do we find the matrix A such that $T(\mathbf{v}) = A\mathbf{v}$?

Thought Experiment

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Suppose I am holding a matrix A

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Your objective is to figure out what A is

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But you're only allowed to ask the question:

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What is $A\mathbf{v}$?

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(you pick the $\mathbf{v}$'s, and I have to tell the truth)

Thought Experiment

Suppose I am holding a matrix A

Your objective is to figure out what A is

But you're only allowed to ask the question:

What is $A\mathbf{v}$?

(you pick the $\mathbf{v}$'s, and I have to tell the truth)

This is basically linear algebraic battleship

Recall: Matrix-Vector Multiplication

$$\left[\vec{a}_1 \quad \dots \quad \vec{a}_n \right] \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix} = v_1 \vec{a}_1 + \dots + v_n \vec{a}_n$$

A

$$\left[\vec{a}_1 \quad \dots \quad \vec{a}_n \right] \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} =$$

$$A \vec{e}_1 = \vec{a}_1$$

$$1 \vec{a}_1 + 0 \vec{a}_2 + \dots + 0 \vec{a}_n$$

first col of A

Recall: Calculating $A\mathbf{v}$

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} = \begin{bmatrix} ? \\ ? \\ \vdots \\ ? \end{bmatrix}$$

$$b_1 = a_{11}v_1 + a_{12}v_2 + \cdots + a_{1n}v_n = \sum_{i=1}^n a_{1i}v_i$$

Recall: Matrix-Vector Multiplication

Definition. Given a $(m \times n)$ matrix A with columns $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n$, and a vector $\mathbf{v}$ in $\mathbb{R}^n$, we define

$$A\mathbf{v} = [\mathbf{a}_1 \quad \mathbf{a}_2 \quad \dots \quad \mathbf{a}_n] \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} = v_1\mathbf{a}_1 + v_2\mathbf{a}_2 + \dots v_n\mathbf{a}_n$$

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$A\mathbf{v}$ is a linear combination of the columns of A with weights given by $\mathbf{v}$

Isolating the first column

The Takeaway

We can learn the first column of the matrix

implementing T by looking at $T \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$

$$T \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 0 \end{pmatrix} = \text{second column of } A \text{ s.r.}$$
$$T(\vec{r}) = A\vec{r}$$

Standard Basis

$$e_i = \begin{bmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{bmatrix}$$

$\begin{matrix} 1 & 2 & \dots & n \end{matrix}$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$e_1 \quad e_2 \quad e_3$

Standard Basis

Df. The *n-dimensional standard basis vectors* (or standard coordinate vectors) are the vectors $\mathbf{e}_1, \dots, \mathbf{e}_n$ where

$$\mathbf{e}_i = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \\ 0 \end{bmatrix} \begin{matrix} 1 \\ 2 \\ \vdots \\ i-1 \\ i \\ i+1 \\ \vdots \\ n-1 \\ n \end{matrix}$$

Standard Basis

Def (Alternative). The n -dimensional standard basis vectors $\mathbf{e}_1, \dots, \mathbf{e}_n$ are the columns of the $n \times n$ identity matrix

$$I = [\mathbf{e}_1 \quad \mathbf{e}_2 \quad \dots \quad \mathbf{e}_n]$$

Standard Basis and the Matrix Equation

$$[\mathbf{a}_1 \quad \mathbf{a}_2 \quad \dots \quad \mathbf{a}_n] \mathbf{e}_i = \mathbf{a}_i$$

$$\sum_{j=1}^n \vec{a}_j (\vec{e}_i)_j = \vec{a}_i$$

$$(\vec{e}_i)_j = \begin{cases} 1 & j=i \\ 0 & \text{o.w.} \end{cases}$$

Standard Basis and Vector Coordinates

$$\begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} = v_1 \mathbf{e}_1 + v_2 \mathbf{e}_2 + \dots + v_n \mathbf{e}_n$$

Column vectors describe how to write a vector as a linear combination of the standard basis

Standard Basis and Linear Transformations

Thm. For a linear transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$, the matrix

$$A = [T(\mathbf{e}_1) \quad T(\mathbf{e}_2) \quad \dots \quad T(\mathbf{e}_n)]$$

is the unique matrix such that $T(\mathbf{v}) = A\mathbf{v}$ for all $\mathbf{v}$ in $\mathbb{R}^n$

Formally

$$T(\mathbf{v}) = \begin{bmatrix} T(\mathbf{e}_1) & T(\mathbf{e}_2) & \dots & T(\mathbf{e}_n) \end{bmatrix} \mathbf{v}$$

$$T(\vec{v}) =$$

$$T\left(\sum_{i=1}^n v_i \vec{e}_i\right) = \text{(by linearity)}$$

$$\sum_{i=1}^n v_i T(\vec{e}_i) = \text{(by def. of mat. rep. m.v.)}$$

$$\begin{bmatrix} T(\vec{e}_1) & \dots & T(\vec{e}_n) \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} = \boxed{\begin{bmatrix} T(\vec{e}_1) & \dots & T(\vec{e}_n) \end{bmatrix} \vec{v}}$$

How To: Matrices of Linear Transformations

Q. Find the matrix that implements the transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$

A. Determine the images of standard basis vectors under T . The desired matrix is

$$\begin{bmatrix} T(\mathbf{e}_1) & T(\mathbf{e}_2) & \dots & T(\mathbf{e}_n) \end{bmatrix}$$

Example

$$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \mapsto \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \mapsto \begin{bmatrix} x_1 + 2x_2 - x_3 \\ 2x_3 \end{bmatrix}$$

$$\vec{x} \mapsto \begin{bmatrix} 1 & 2 & -1 \\ 0 & 0 & 2 \end{bmatrix} \vec{x}$$

same

equivalent

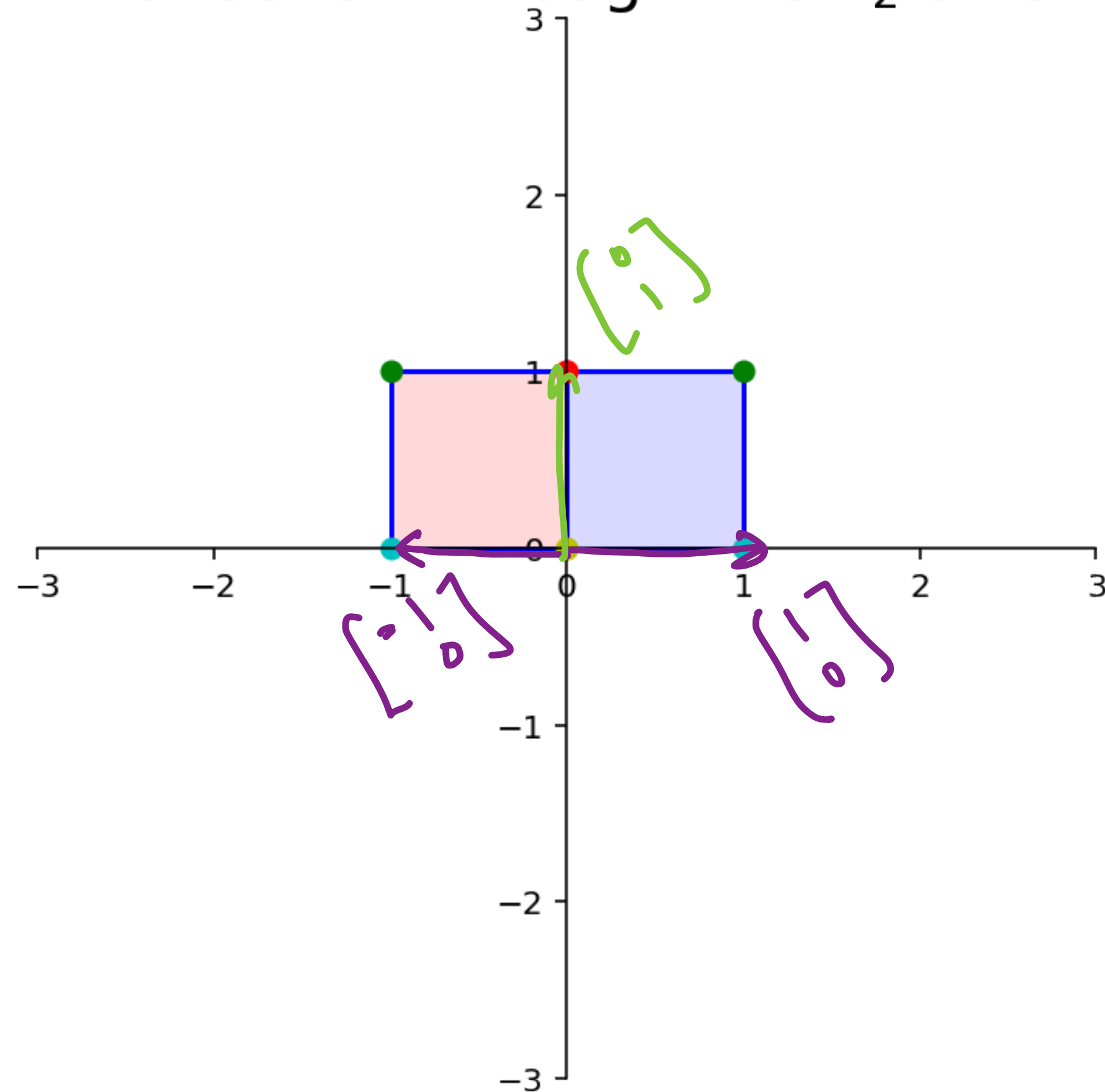
$$\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \mapsto \begin{bmatrix} 0 + 2(1) - 0 \\ 2(0) \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & -1 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_1 + 2x_2 - x_3 \\ 2x_3 \end{bmatrix}$$

$$\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \mapsto \begin{bmatrix} 0 + 2(0) - 1 \\ 2(1) \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

Example: Reflection through the x_2 -axis

Reflection through the x_2 axis



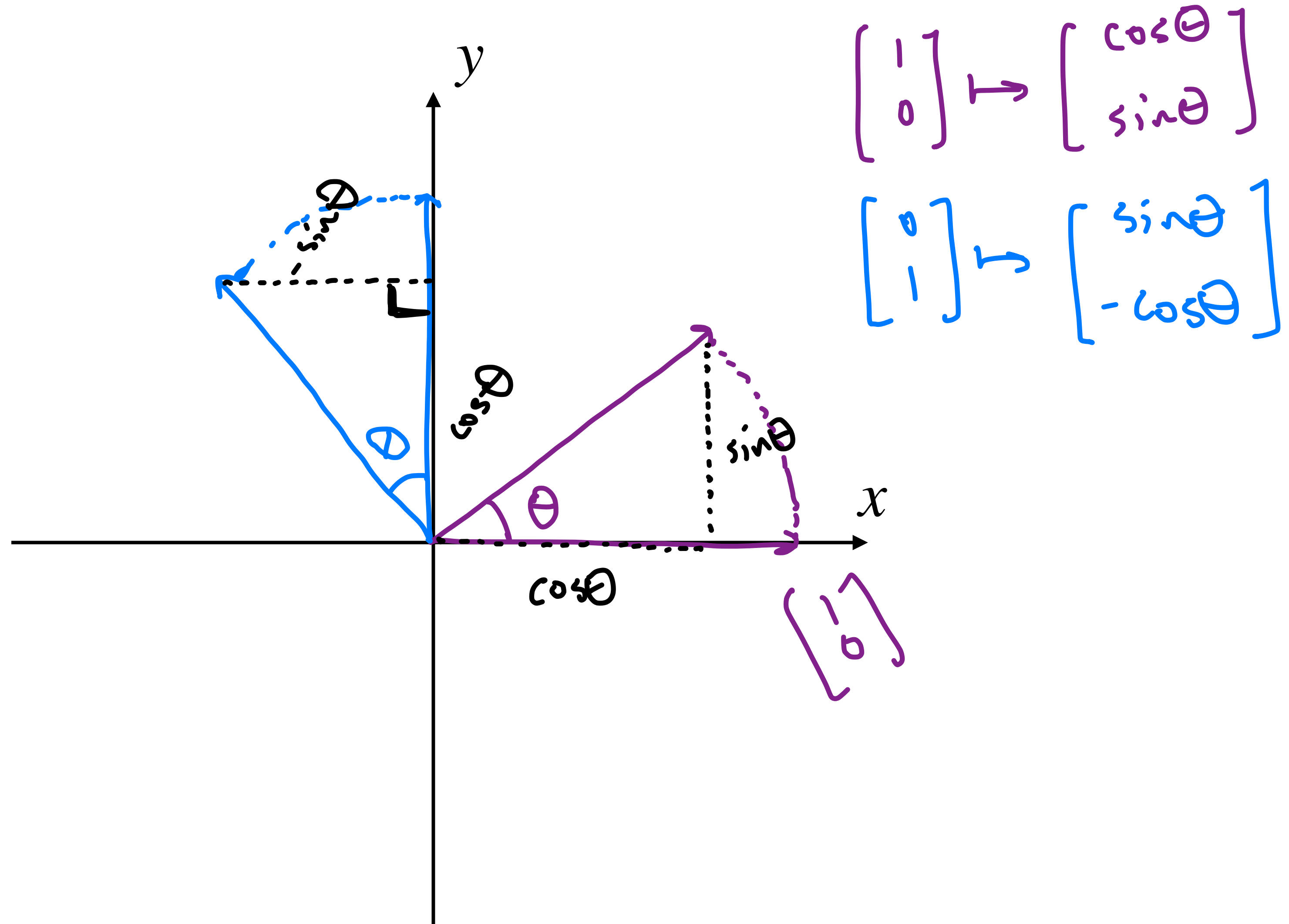
$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} \mapsto \begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 \\ 1 \end{bmatrix} \mapsto \begin{bmatrix} 0 \\ -1 \end{bmatrix}$$

$$\vec{x} \mapsto \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \vec{x}$$

Example: General Rotation

How does rotation affect the standard basis?



Rotation Matrix

$$\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

The Takeaway: We can figure out the matrix that implements a linear transformations by understanding what it does to the standard basis

One-to-One and Onto

Recall: Motivating Questions

What kind of functions can we define in this way?

How do we interpret what the transformation does to a set of vectors?

How does this relate back to matrix equations?

Recall: A New Interpretation of the Matrix Equation

$A\mathbf{x} = \mathbf{b}?$ $\equiv$ is there a vector which A
transforms into $\mathbf{b}$?

Solve $A\mathbf{x} = \mathbf{b}$ $\equiv$ find a vector which A
transforms into $\mathbf{b}$

Recall: A New Interpretation of the Matrix Equation

$A\mathbf{x} = \mathbf{b}?$ $\equiv$ is there a vector which A
transforms into $\mathbf{b}$?

Solve $A\mathbf{x} = \mathbf{b}$ $\equiv$ find a vector which A
transforms into $\mathbf{b}$

What about other questions?

Other Questions Like...

(full span)

Does $A\mathbf{x} = \mathbf{b}$ have a solution for any choice of $\mathbf{b}$?

Does $A\mathbf{x} = \mathbf{0}$ have a unique solution? (lin. ind.)

Other Questions Like...

Does $A\mathbf{x} = \mathbf{b}$ have at least one solution for any choice of $\mathbf{b}$?

Does $A\mathbf{x} = \mathbf{b}$ have at most one solution for any choice of $\mathbf{b}$?

Verify

$A\mathbf{x} = \mathbf{0}$ has a unique solution $\equiv A\mathbf{x} = \mathbf{b}$ has at most one solution

$(\Rightarrow)$ Suppose $A\vec{v}_1 = \vec{b}$ $A\vec{v}_2 = \vec{b}$ $\vec{v}_1 \neq \vec{v}_2$

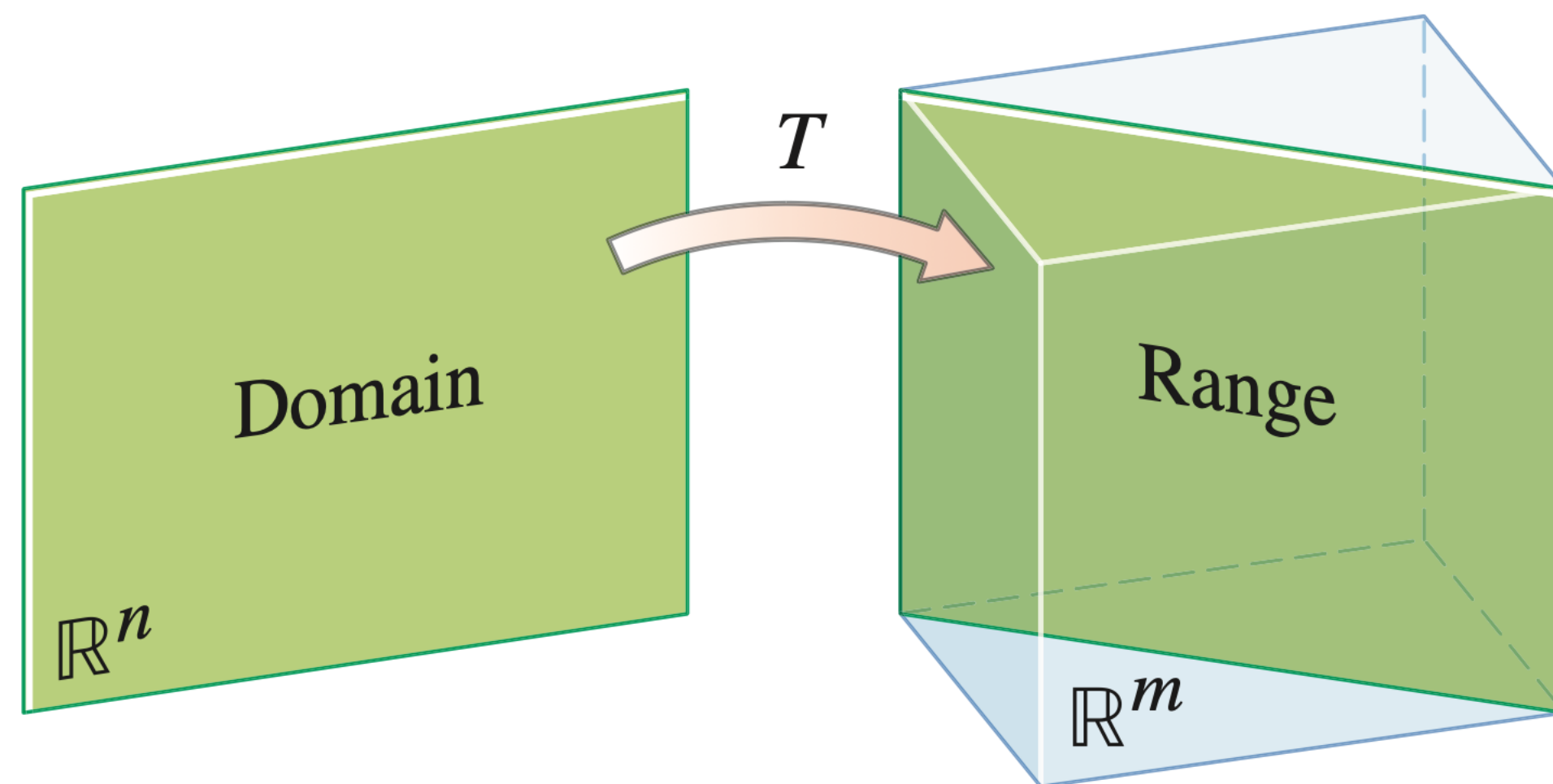
$$A(\vec{v}_1 - \vec{v}_2) = A\vec{v}_1 - A\vec{v}_2 = \vec{b} - \vec{b} = \vec{0}$$

nonzero, nontrivial solution

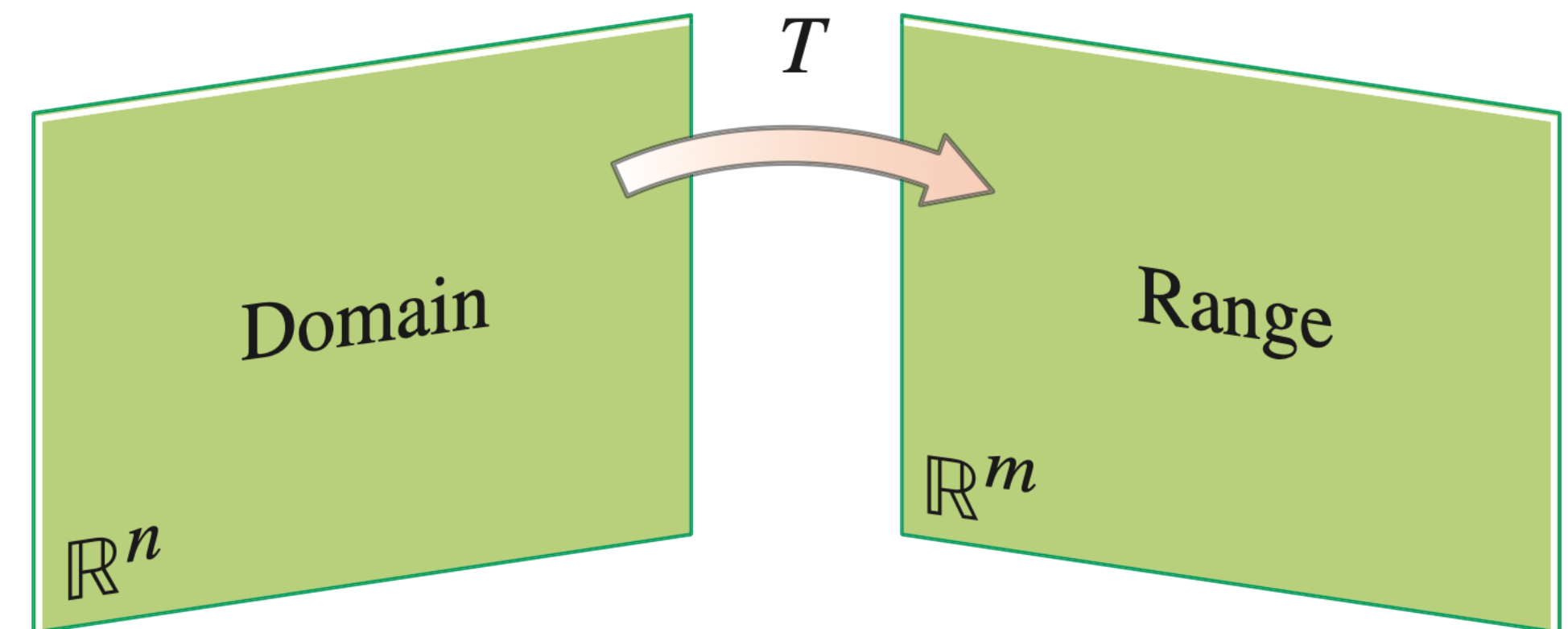
$(\Leftarrow)$ Challenge

Onto

Onto (Pictorially)

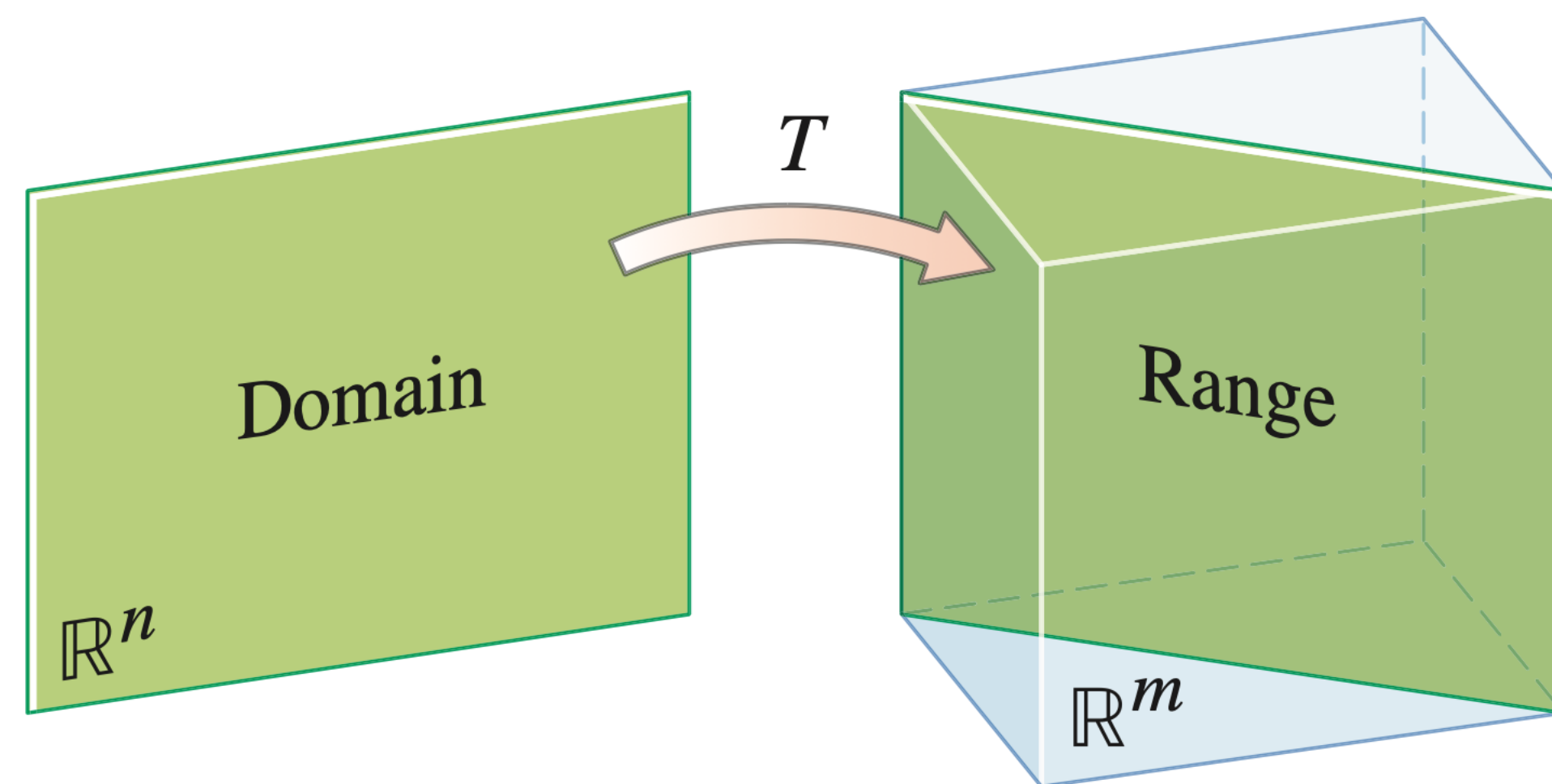


T is *not* onto $\mathbb{R}^m$

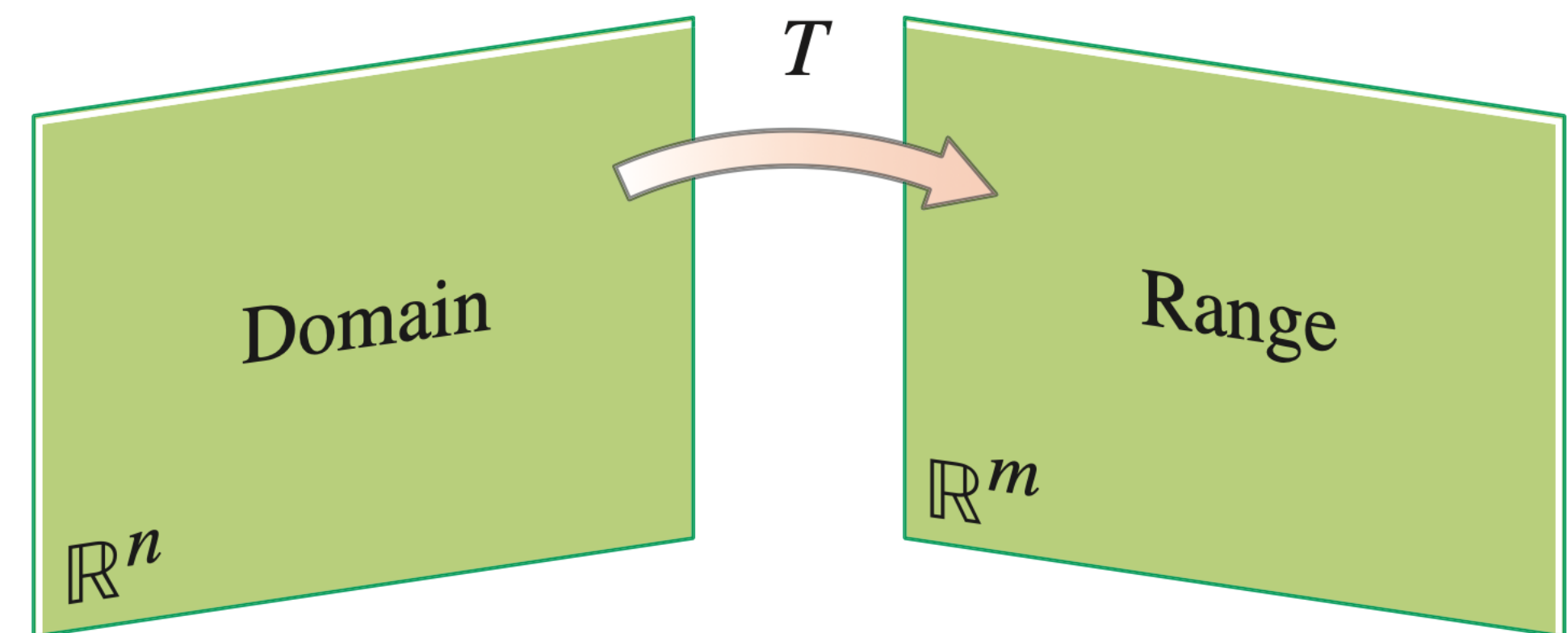


T is onto $\mathbb{R}^m$

Onto (Pictorially)



T is *not* onto $\mathbb{R}^m$

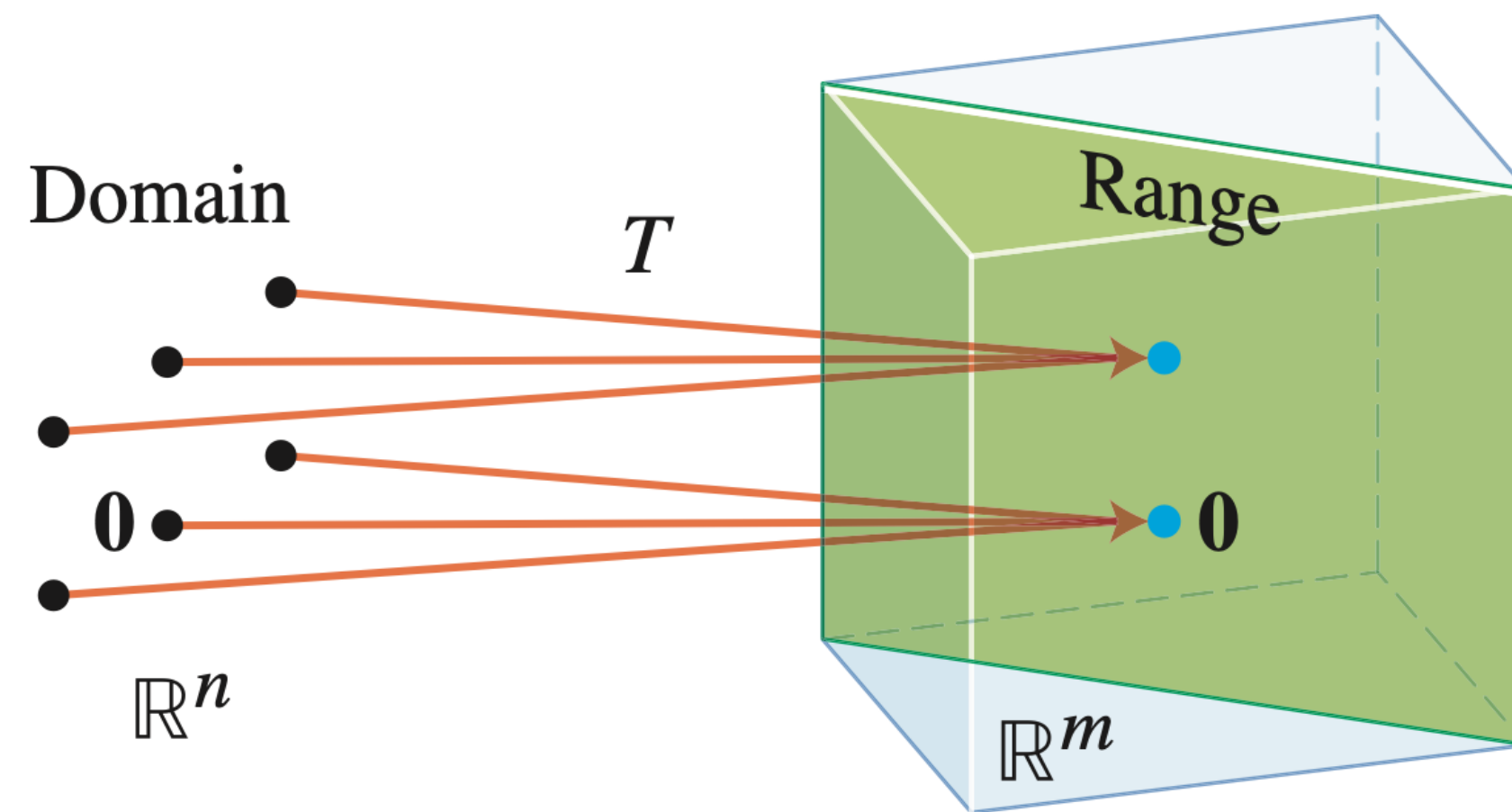


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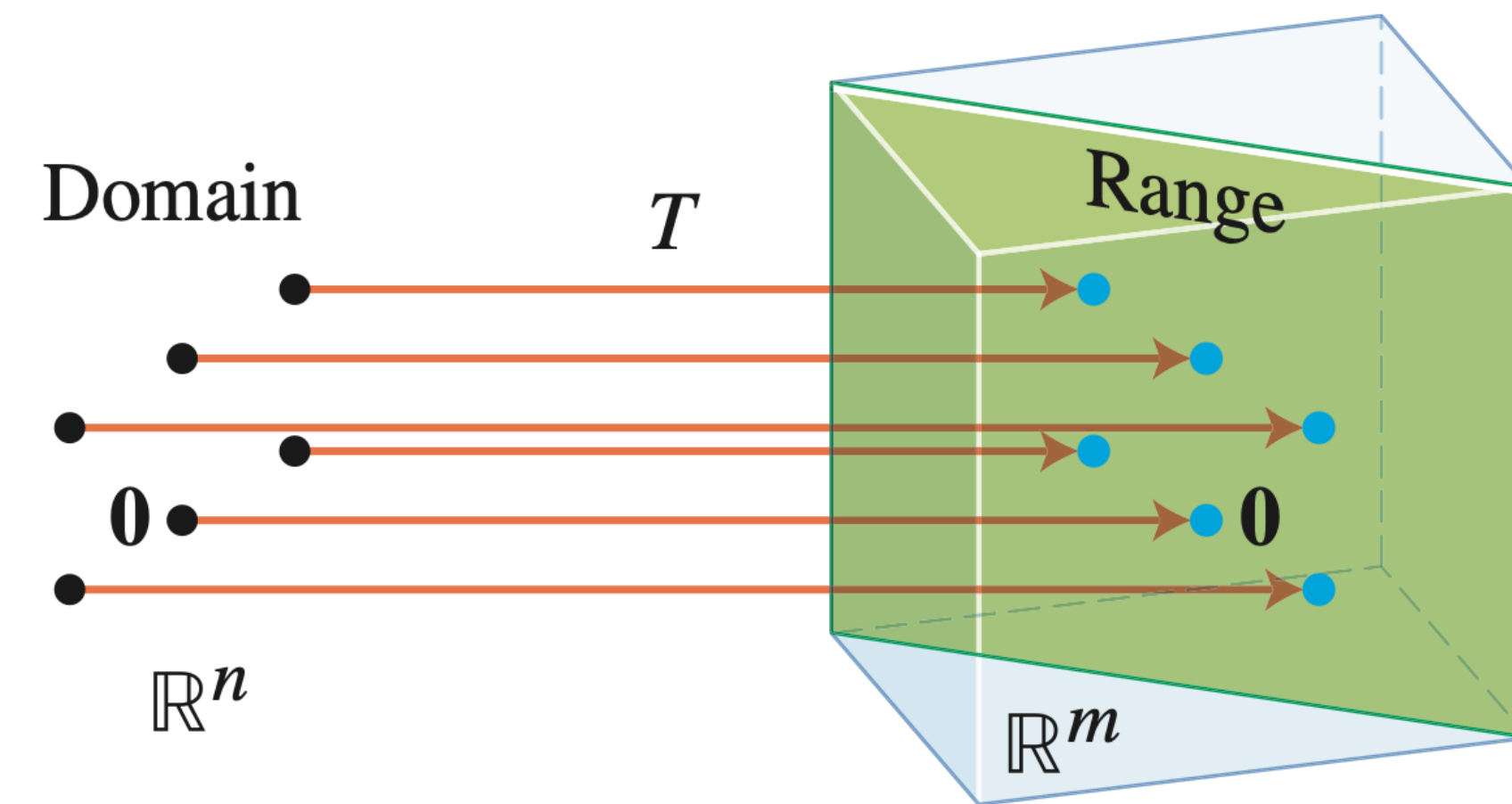
T is onto if its range = its codomain

One-to-One

One-to-One (Pictorially)



T is *not* one-to-one

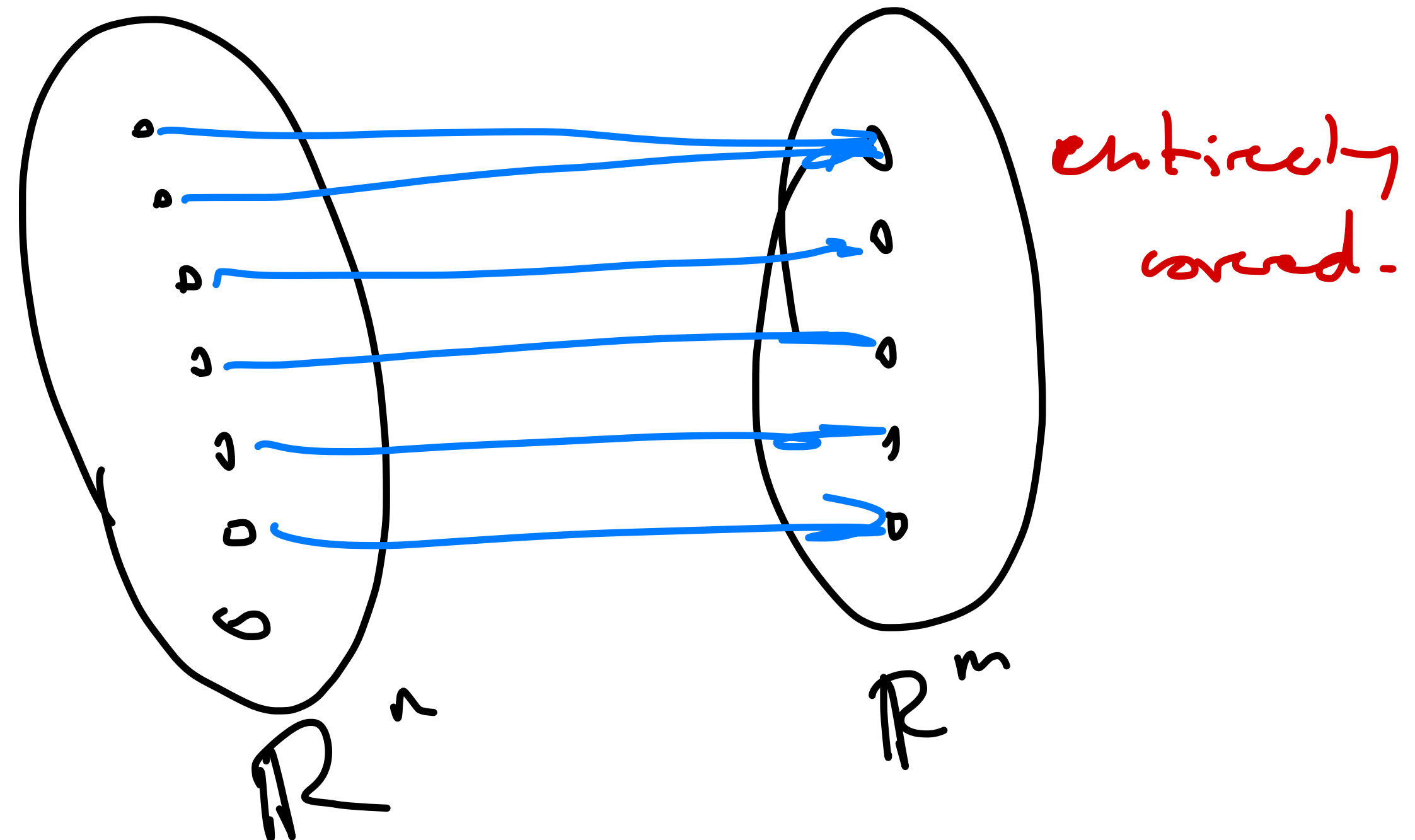


T is one-to-one

Onto and One-to-One

Onto and One-to-One

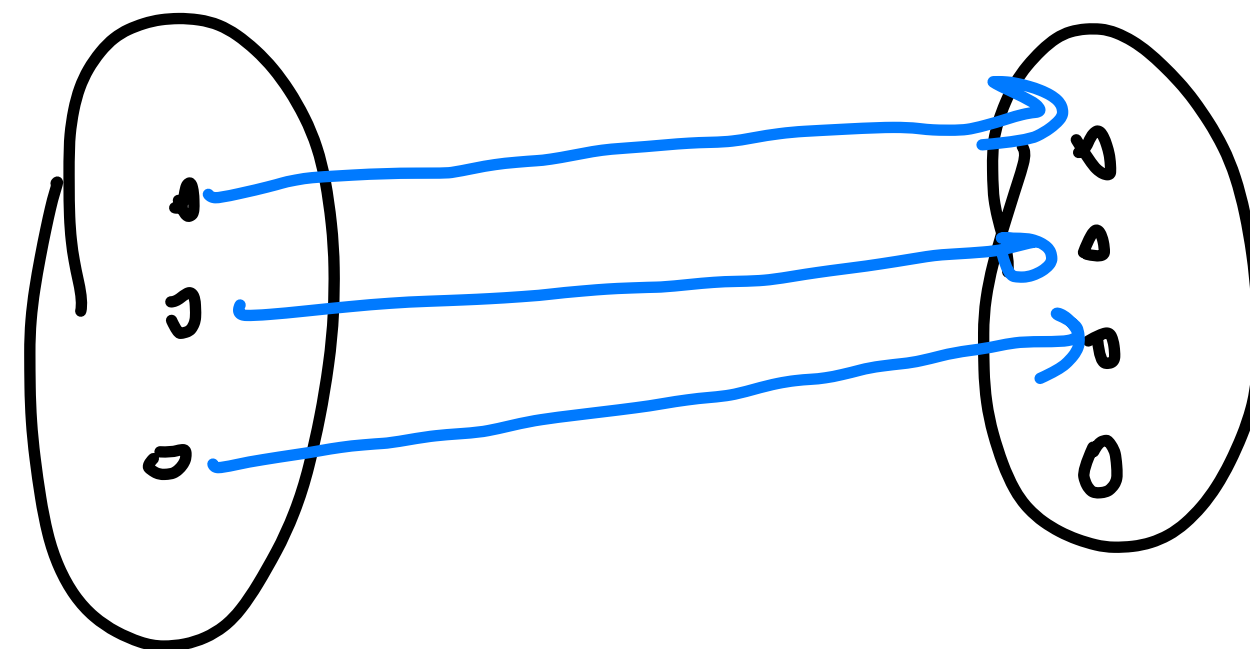
Df. $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is **onto** if any $\mathbf{b} \in \mathbb{R}^m$ is the **image** of at least one vector $\mathbf{v} \in \mathbb{R}^n$ (i.e., $T(\mathbf{v}) = \mathbf{b}$)



Onto and One-to-One

Df. $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is **onto** if any $\mathbf{b} \in \mathbb{R}^m$ is the **image of at least one vector** $\mathbf{v} \in \mathbb{R}^n$ (i.e., $T(\mathbf{v}) = \mathbf{b}$)

Df. $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is **one-to-one** if any $\mathbf{b} \in \mathbb{R}^m$ is the **image of at most one vector** $\mathbf{v} \in \mathbb{R}^n$



Taking Stock: Onto

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Thm. The following are logically equivalent for
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» $A\mathbf{x} = \mathbf{b}$ has a solution for any choice of $\mathbf{b}$

Taking Stock: Onto

Thm. The following are logically equivalent for $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ implemented by A

- » T is onto
- » $A\mathbf{x} = \mathbf{b}$ has a solution for any choice of $\mathbf{b}$
- » $\text{range}(T) = \text{codomain}(T)$

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- » the columns of A span $\mathbb{R}^m$
- » A has a pivot position in every row

Taking Stock: One-to-One

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Theorem. The following are logically equivalent
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Theorem. The following are logically equivalent for $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ implemented by A

» T is one-to-one

» $A\mathbf{x} = \mathbf{b}$ has at most one solution for any $\mathbf{b}$

Taking Stock: One-to-One

Theorem. The following are logically equivalent for $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ implemented by A

- » T is one-to-one
- » $A\mathbf{x} = \mathbf{b}$ has at most one solution for any $\mathbf{b}$
- » $A\mathbf{x} = \mathbf{0}$ has only the trivial solution

Taking Stock: One-to-One

Theorem. The following are logically equivalent for $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ implemented by A

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- » The columns of A are linearly independent

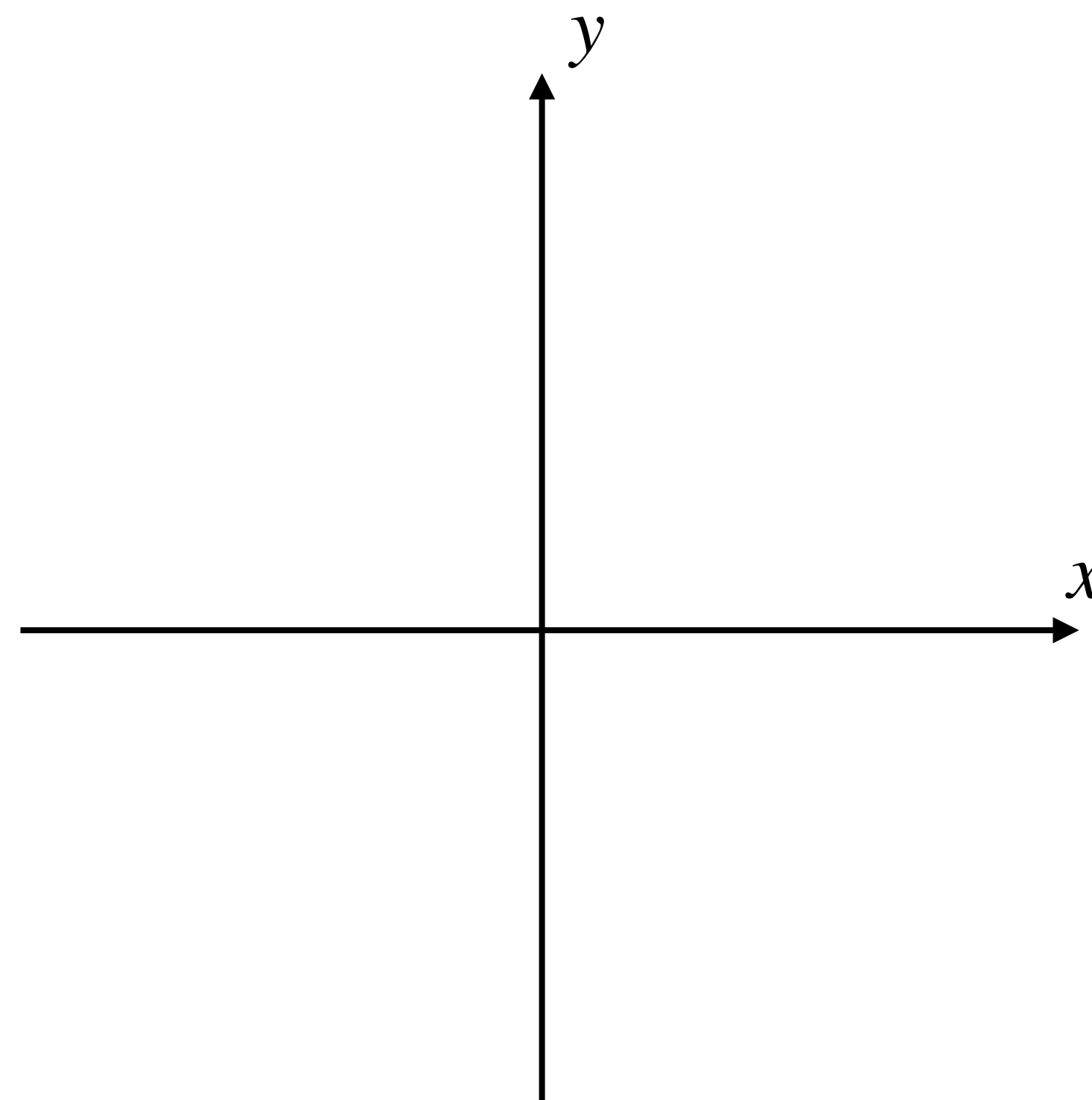
Taking Stock: One-to-One

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- » The columns of A are linearly independent
- » A has a pivot position in every column

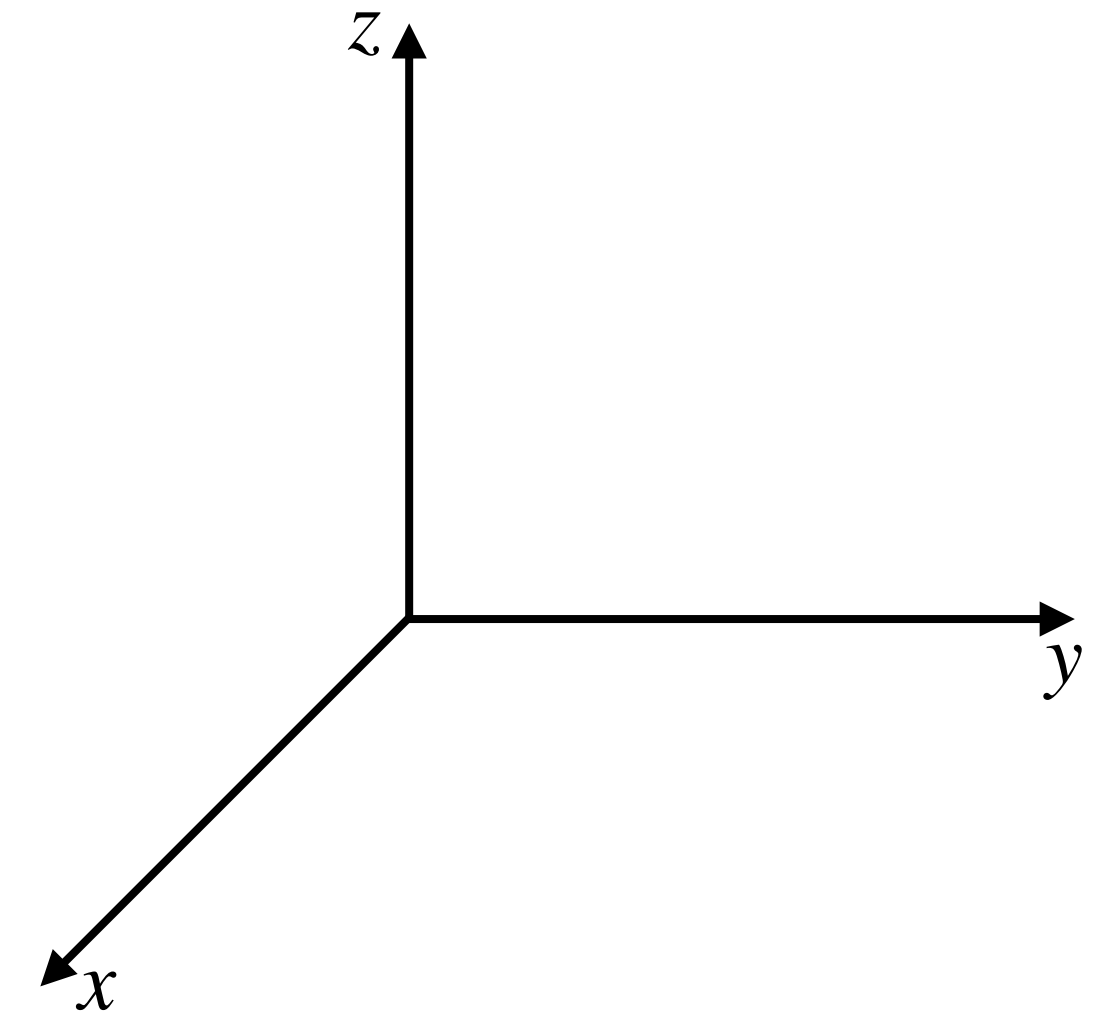
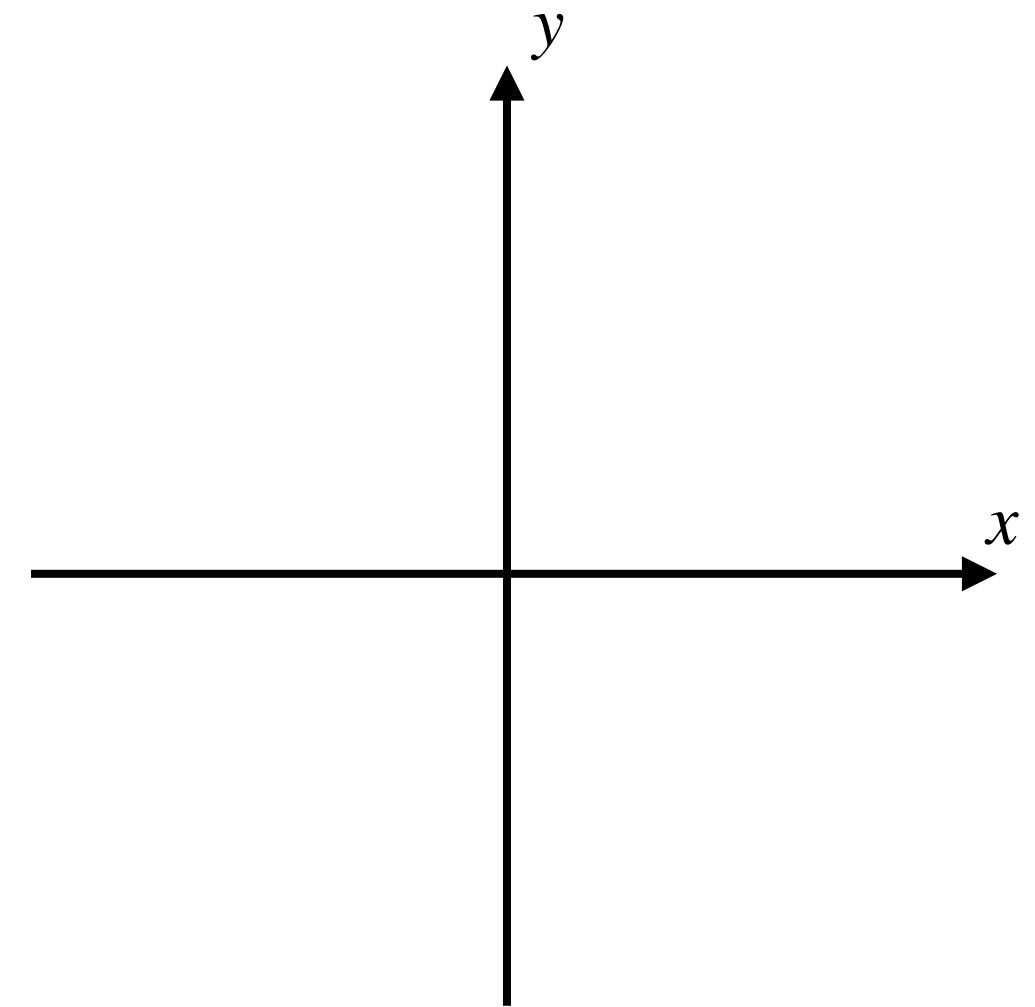
Example: both 1-1 and onto

$$\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$



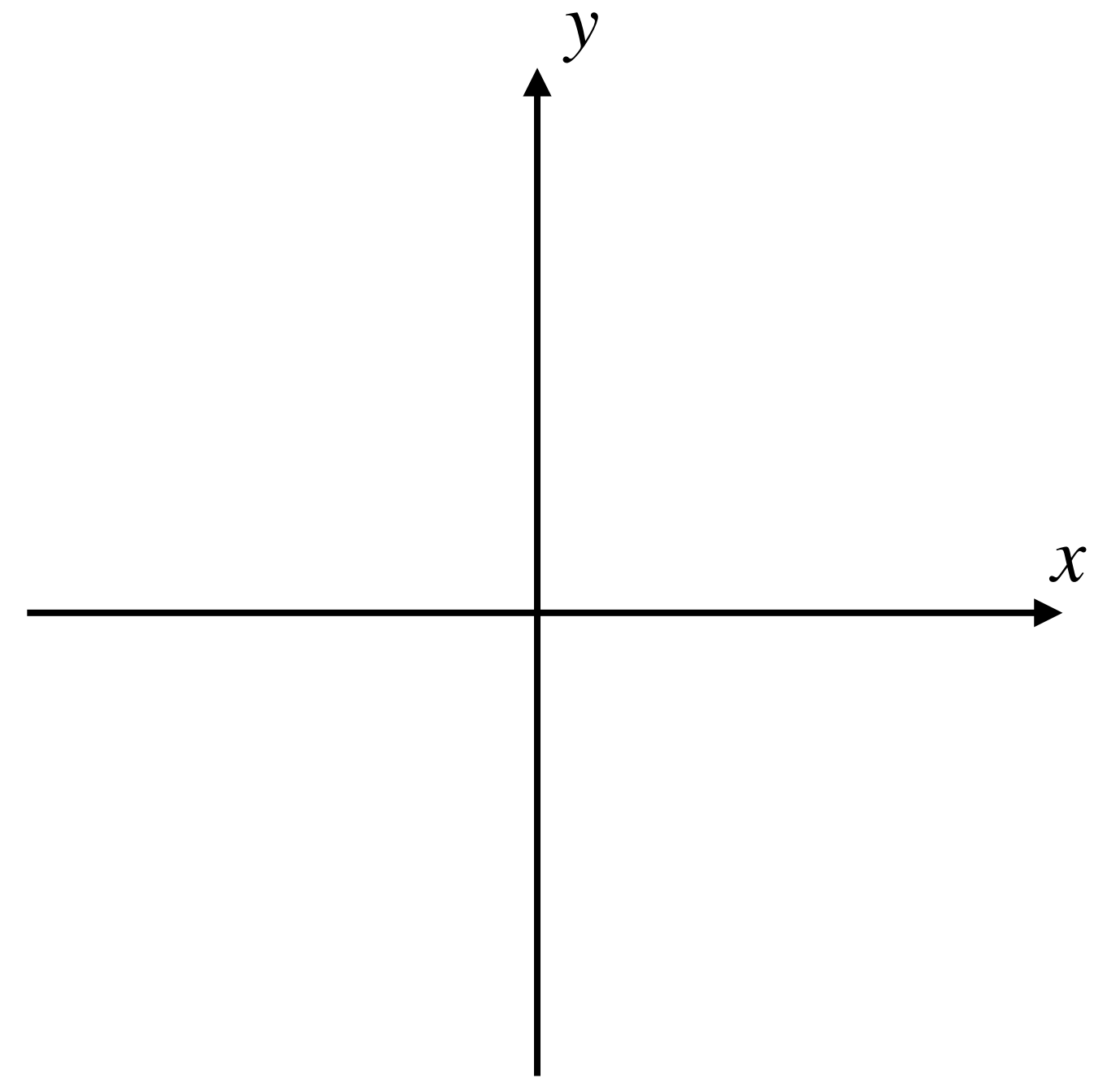
Example: 1-1, not onto

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \mapsto \begin{bmatrix} x_1 \\ x_2 \\ x_1 + x_2 \end{bmatrix}$$



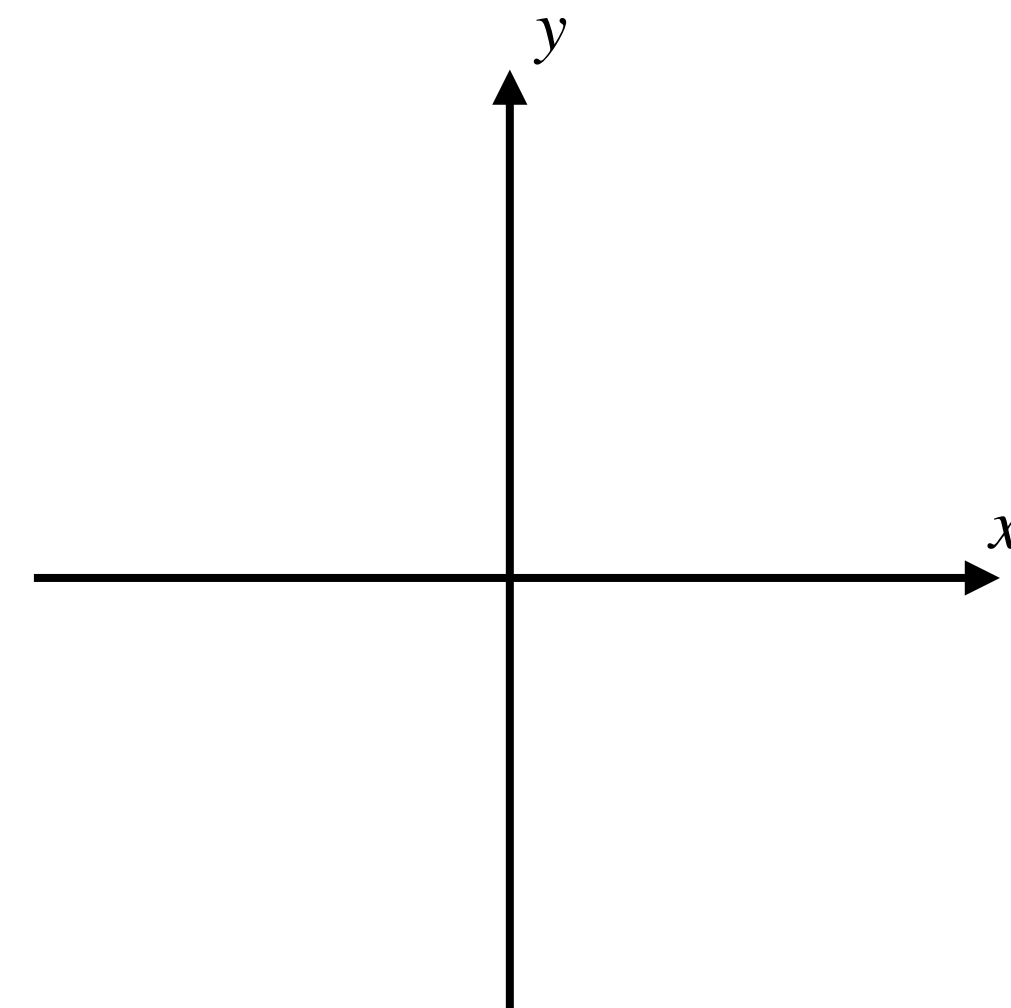
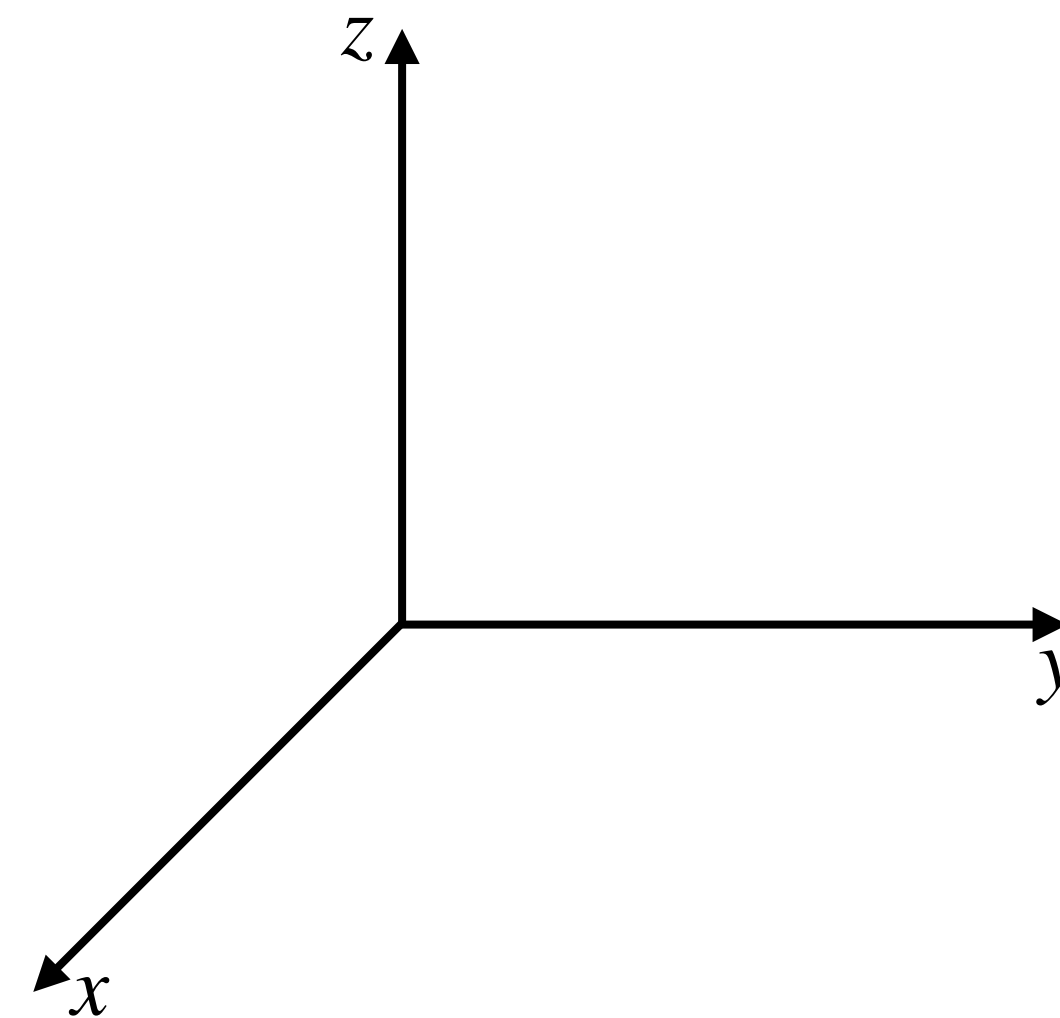
Example: not 1-1, not onto

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$



Example: onto, not 1-1

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \mapsto \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$



Summary

Matrix transformations and linear transformations are **the same thing**

We can find these matrices by looking at how the transformation behaves on the **standard basis**

We can reason about matrix equations by directly reasoning about the linear transformations